Chapter 3

BIPOLAR JUNCTION TRANSISTORS


2N3055
6-52




microelectronics group
A replica of the first transistor, invented at Bell Labs, December 23, 1947
Lucent Technologies
Bell Labs Innovations
50 Years and Counting...

CHAPTER OUTLINE

3-1 Structure of Bipolar Junction Transistors

3-2 BJT Bias Circuit

3-3 Amplifier Concepts and AC Considerations

3-4 Small-Signal Amplifiers

3-5 The Bipolar Transistor as a Switch

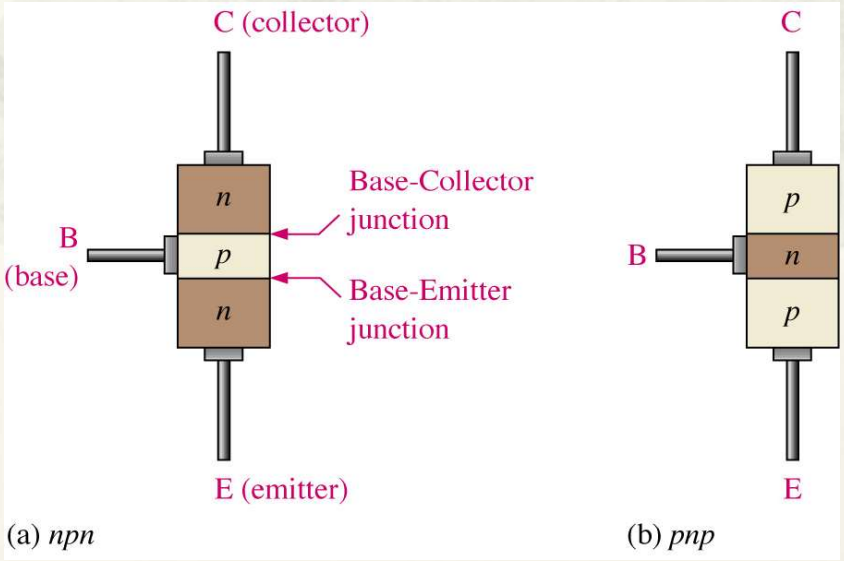
3-6 Transistor Packages and Terminal Identification

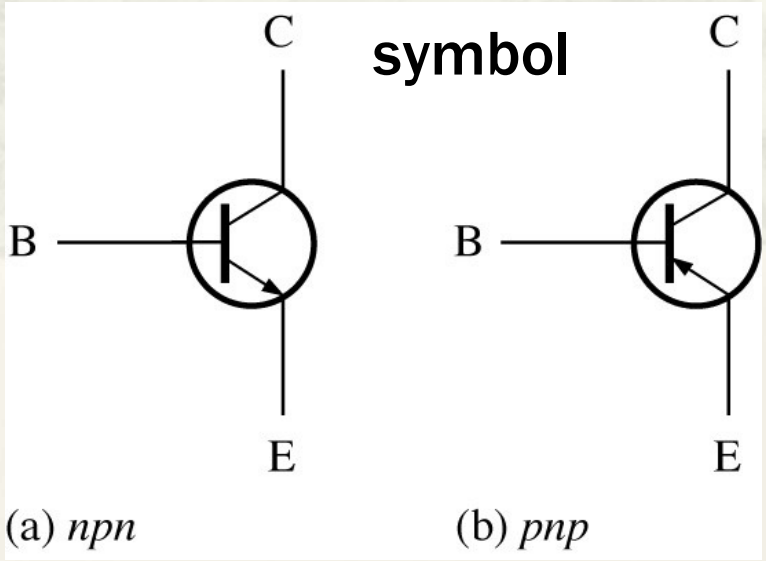
3-1 Structure of Bipolar Junction Transistors

- Transistor Operation
- Transistor Currents
- DC Beta (β_{DC})
- Transistor Voltages
- Characteristic Curve for a BJT
- Cutoff and Saturation
- DC Load Line
- Q-Point
- BJT Parameters

Bipolar Junction Transistor

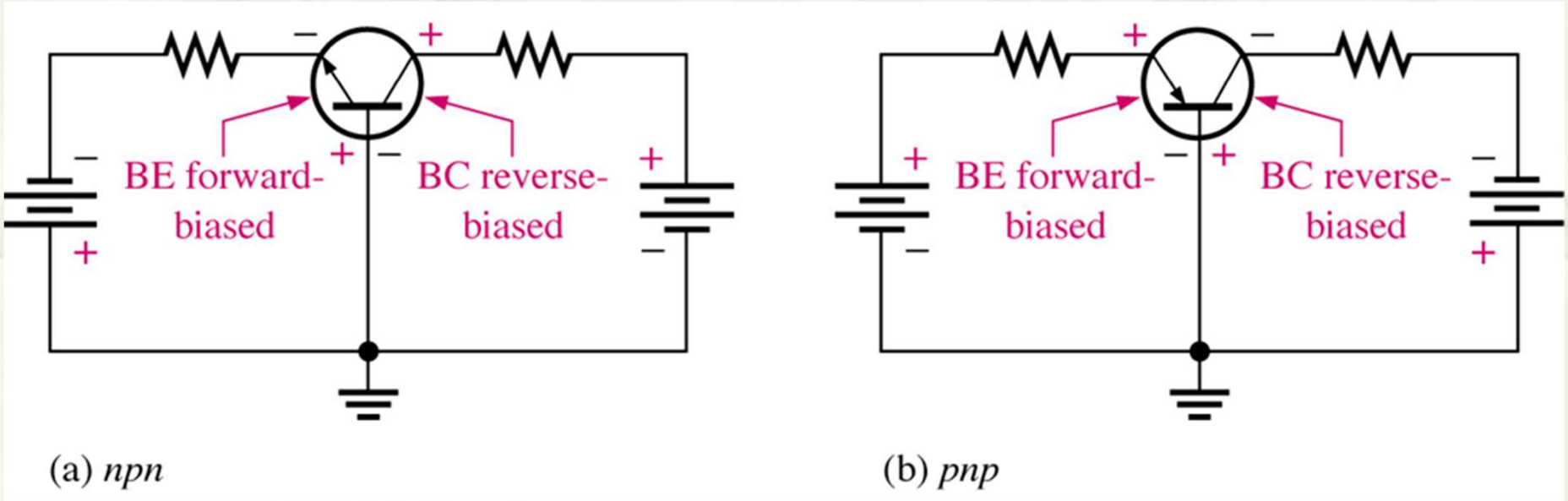
- **Bipolar junction transistor (BJT)**
  - emitter, base, collector
  - base-emitter junction, base-collector junction
  - *npn*, *pnp*
  - bipolar





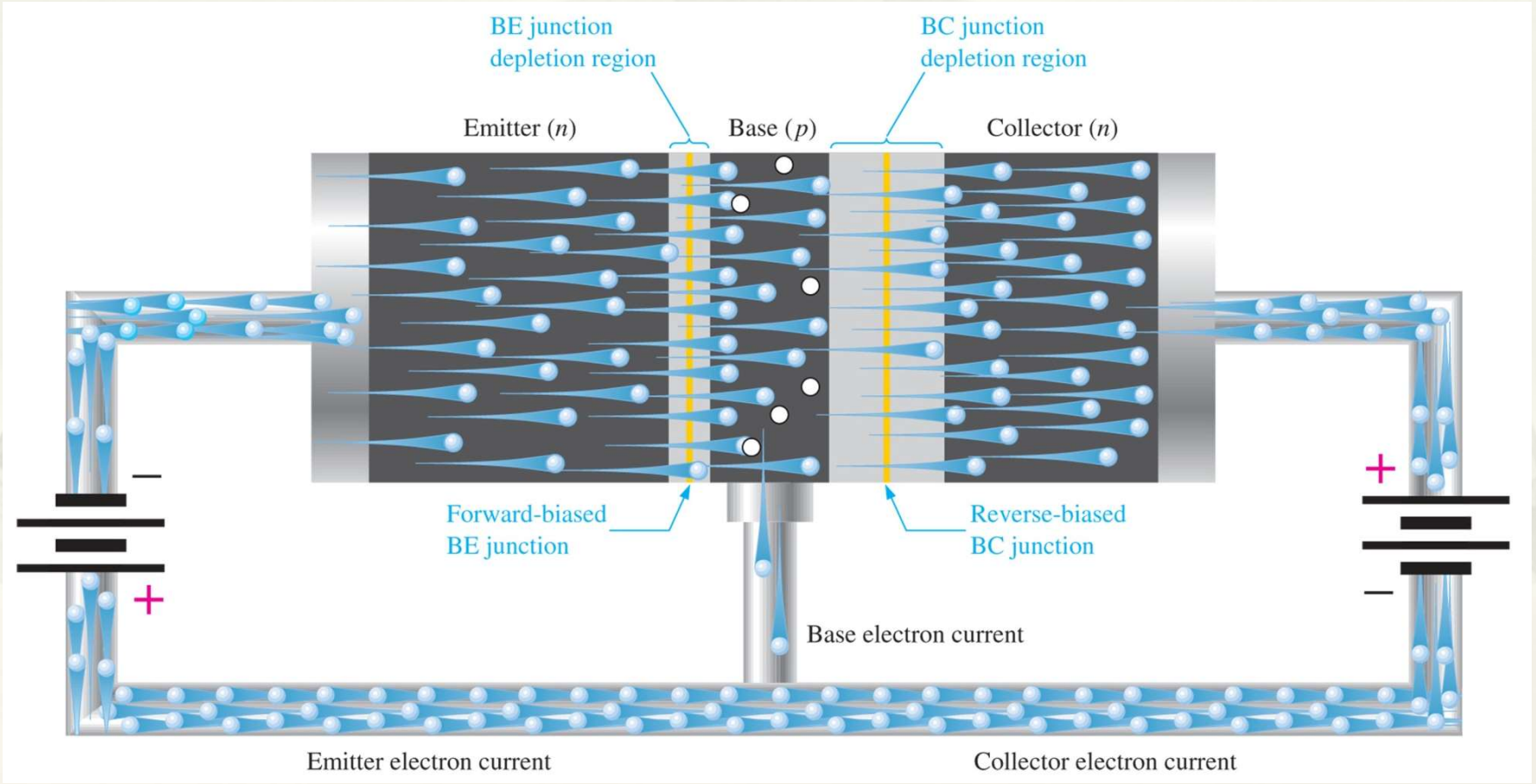
Transistor Operation

- **Forward-reverse bias**
  - The base-emitter (BE) junction is forward-biased and the base-collector (BC) junction is reverse-biased.



Transistor Operation

- **BJT Action**



A small base current can control a larger collector current.

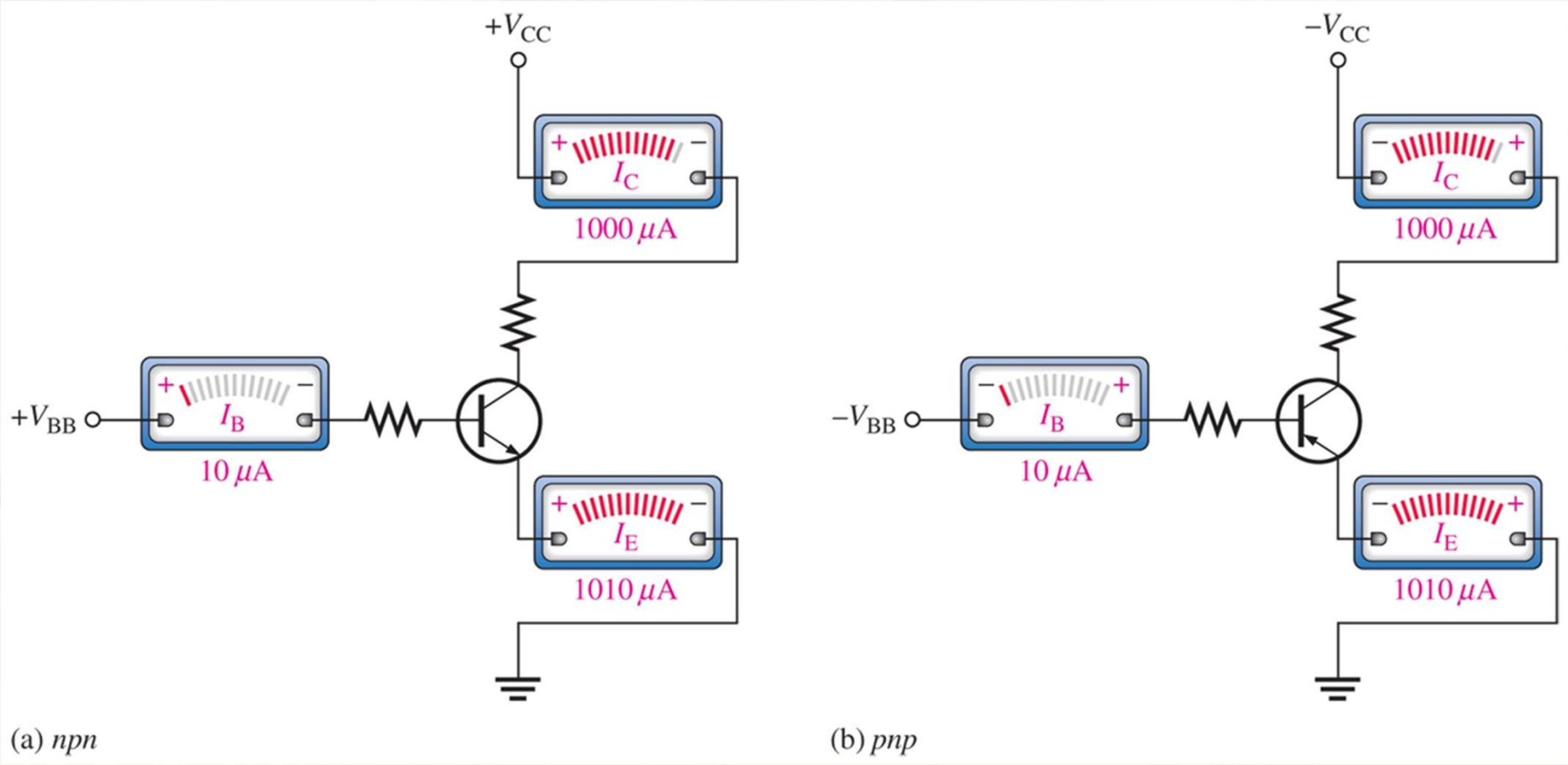
Transistor Currents

- **Kirchhoff's current law (KCL)**
  - $I_E = I_C + I_B$
- **Common-base current gain (α)**
  - $I_C = \alpha I_E$
  - $I_E \cong I_C$
- **Common-emitter current gain (β)**
  - $I_C = \beta I_B$

$$\left.\begin{aligned} I_E &= I_C + I_B \\ I_C &= \alpha I_E \\ I_C &= \beta I_B \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} \alpha &= \frac{\beta}{1+\beta} \\ \beta &= \frac{\alpha}{1-\alpha} \end{aligned}\right.$$

Transistor Currents

- Currents in small-signal transistors



DC Beta (β_{DC})

- **DC Beta (β_{DC})**
  - Current gain of a transistor, is the ratio of the dc collector current to the dc base current.
  - $\beta_{\mathrm{DC}} = \frac{I_{\mathrm{C}}}{I_{\mathrm{B}}}$
  - $h_{\mathbf{FE}}$
  - The value for β_{DC} vary widely and depend on the type of transistor.
  - Typically from 20 (power transistors) to 200 (small-signal transistors)

Transistor Voltage

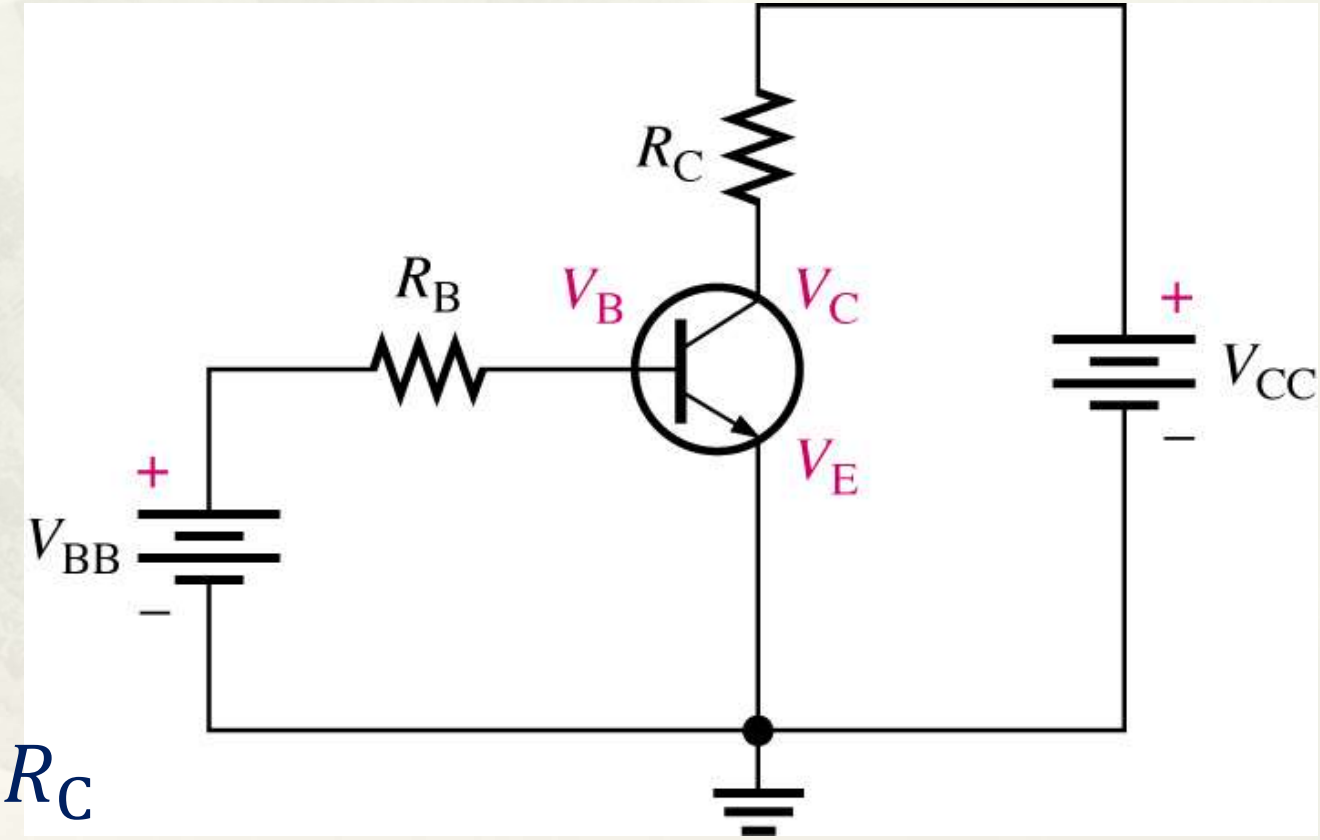
- **Operation condition**

$$V_{BE} \geq V_{th,j}$$

$$V_{BC} < 0$$

$$\Downarrow$$

$$V_C > V_B > V_E$$



- $V_C = V_{CC} - I_C R_C$
- $V_B = V_E + V_{BE} = V_E + 0.7\text{V}$

Example 3-1

Problem

Determine I_B, I_C, I_E, V_B, and V_C in Figure 3–7, where β_{DC} is 48.

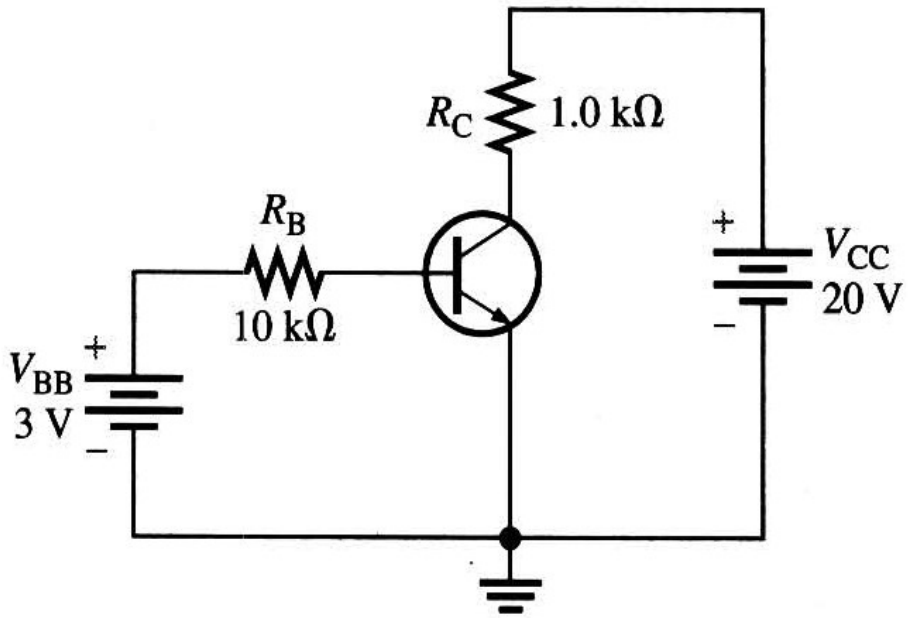


FIGURE 3–7

Example 3-1

Solution

Since V_E is ground, $V_B = \mathbf{0.7\ V}$. The drop across R_B is $V_{BB} - V_B$, so I_B is calculated as follows:

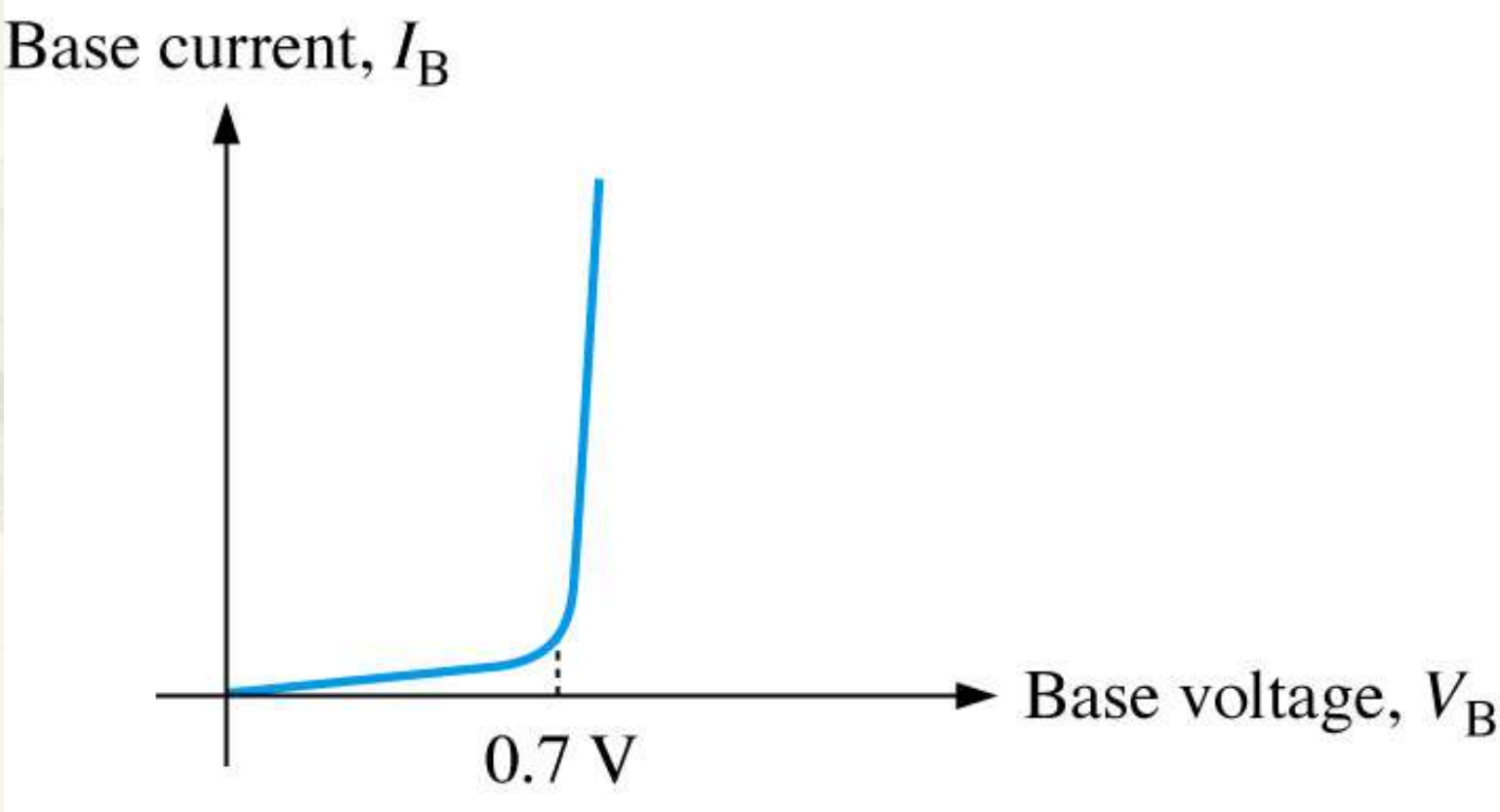
$$I_B = \frac{V_{BB} - V_B}{R_B} = \frac{3\ \text{V} - 0.7\ \text{V}}{10\ \text{k}\Omega} = \mathbf{0.23\ mA}$$

Now you can find I_C, I_E, and V_C.

$$I_C = \beta_{DC} I_B = 48(0.23\ \text{mA}) = \mathbf{11.0\ mA}$$
$$I_E = I_C + I_B = 11.0\ \text{mA} + 0.23\ \text{mA} = \mathbf{11.3\ mA}$$
$$V_C = V_{CC} - I_C R_C = 20\ \text{V} - (11.0\ \text{mA})\ (1.0\ \text{k}\Omega) = \mathbf{9.0\ V}$$

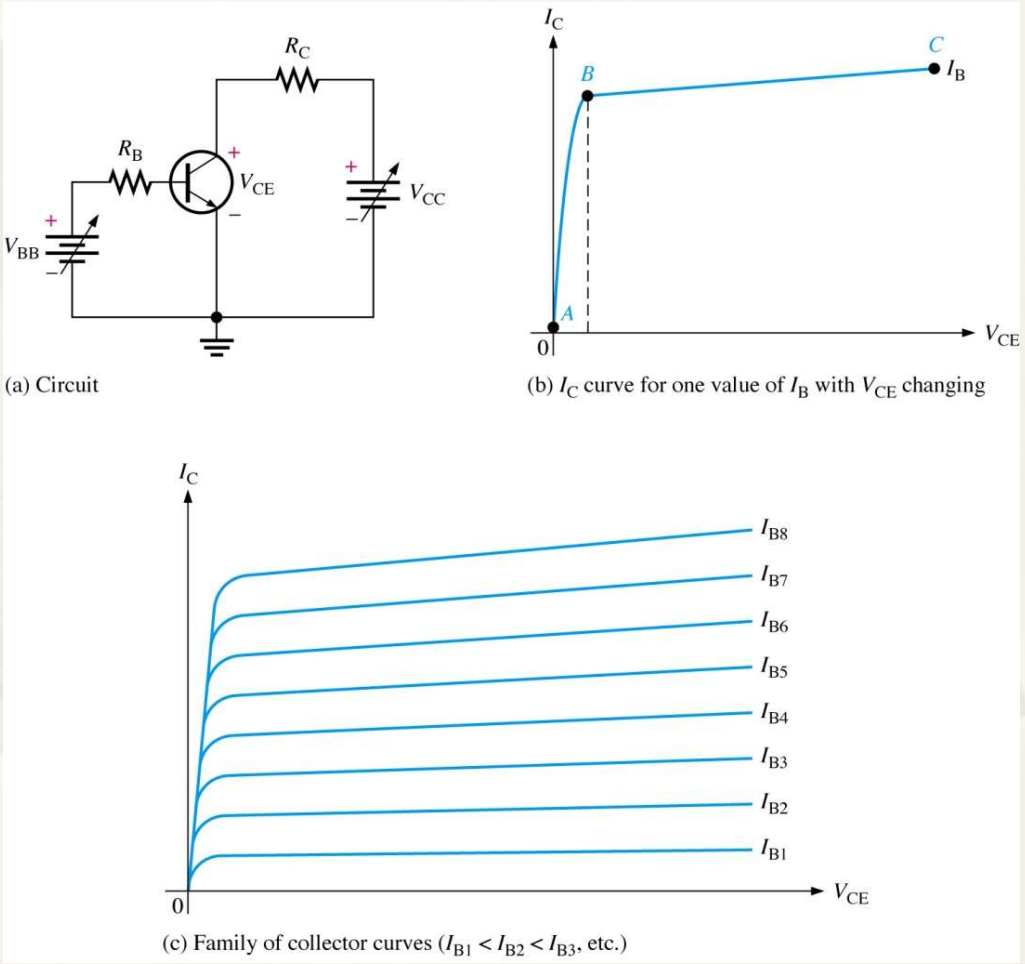
Characteristic Curves for a BJT

- **Base-Emitter Characteristic**
  - It is identical to that of an ordinary diode.



Characteristic Curves for a BJT

- **Collector Characteristic**
  - A family of curves allows you to visualize the complex situation when three variables interact. By holding one of them(I_B) constant, you can see the relationship between the other two ($I_C$versus V_{CE}).
  - Forward Early voltage



Example 3-2

Problem

Sketch the family of collector curves for the circuit in Figure 3–10 for $I_B = 5\ \mu A$ to $25\ \mu A$ in $5\ \mu A$ increments. Assume that $\beta_{DC} = 100$.

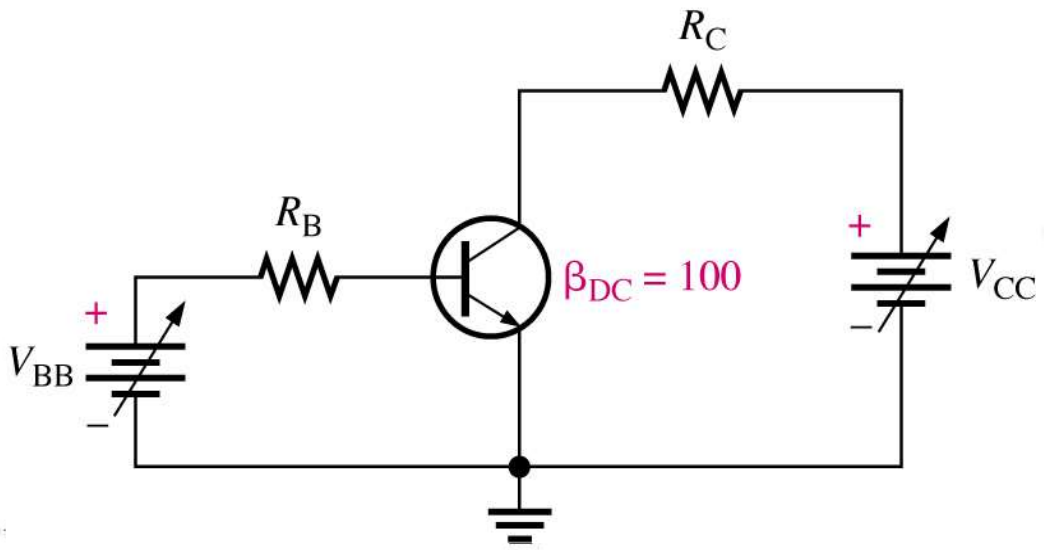


FIGURE 3–10

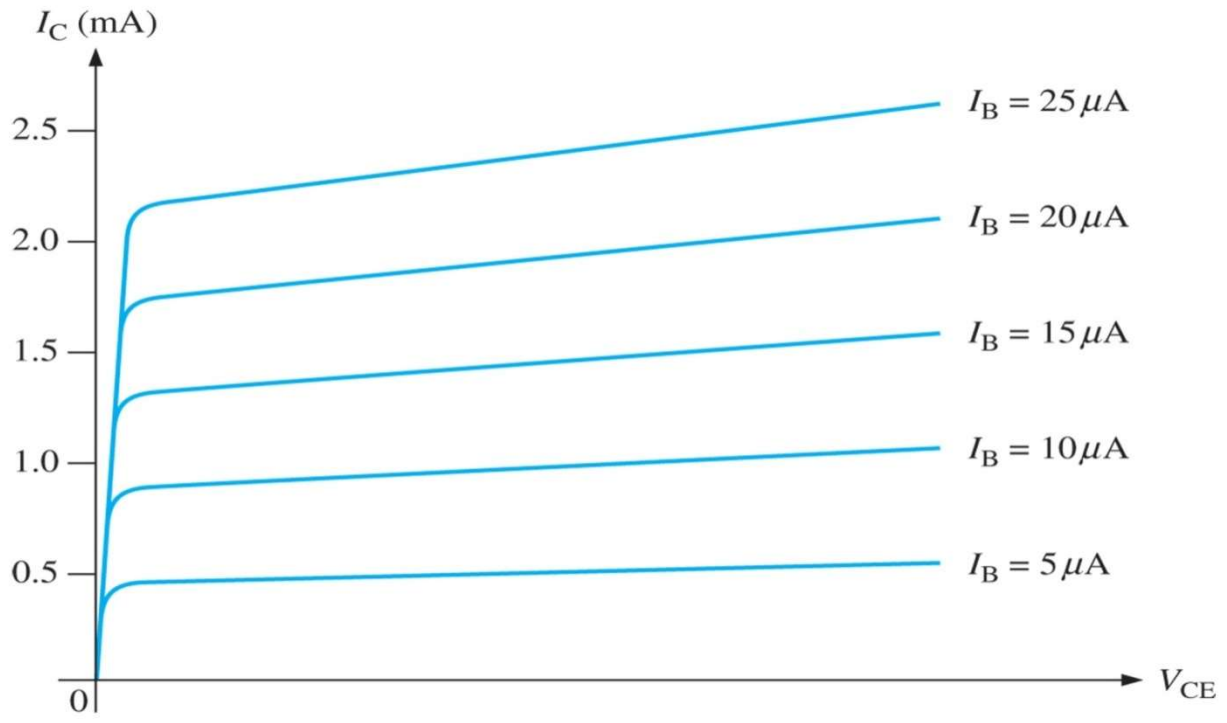
Example 3-2

Solution

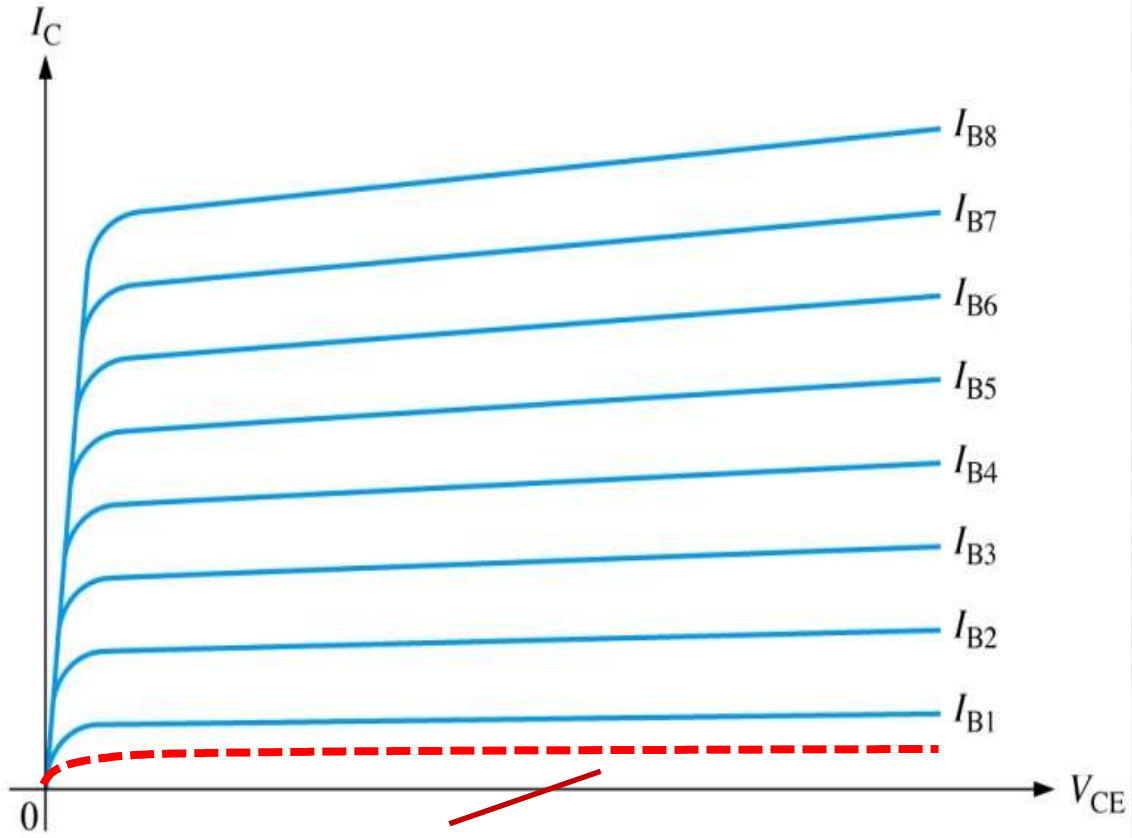
Table 3–1 shows the calculations of I_C using the relationship $I_C = \beta_{DC} I_B$. The resulting curves are plotted in Figure 3–11. To account for the forward Early voltage, the resulting curves are shown with a slight upward slope as previously discussed.

Table 3–1

I_B	I_C
5 μA	0.5 mA
10 μA	1.0 mA
15 μA	1.5 mA
20 μA	2.0 mA
25 μA	2.5 mA

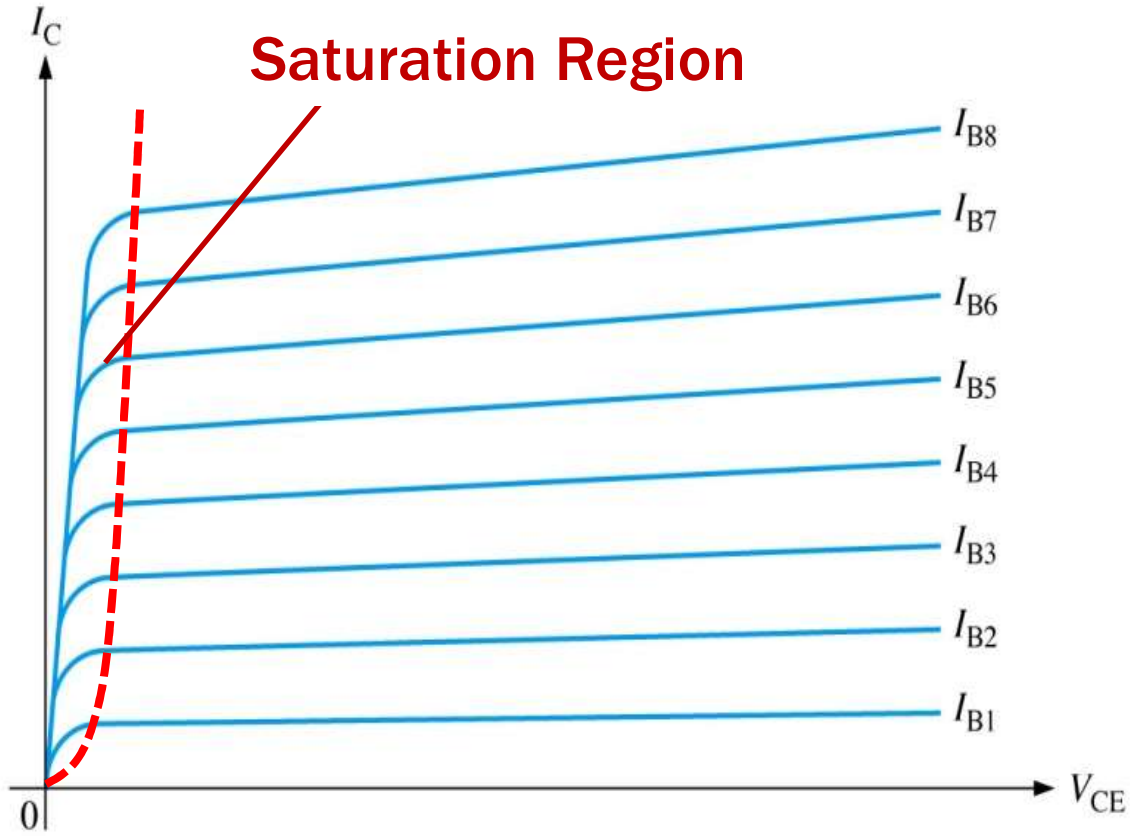


Cutoff and Saturation



- **Cutoff**
  - When $I_B = 0$, the transistor is in cutoff and there is essentially no collector current except for a very tiny amount of collector leakage current, I_{CEO}.
  - In cutoff, both the base-emitter and base-collector junctions are reverse-biased.

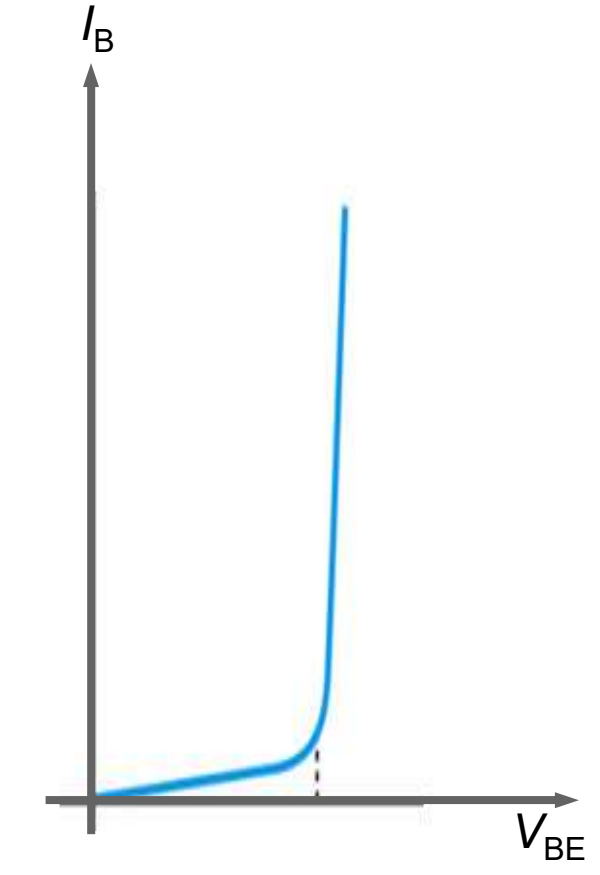
Cutoff and Saturation

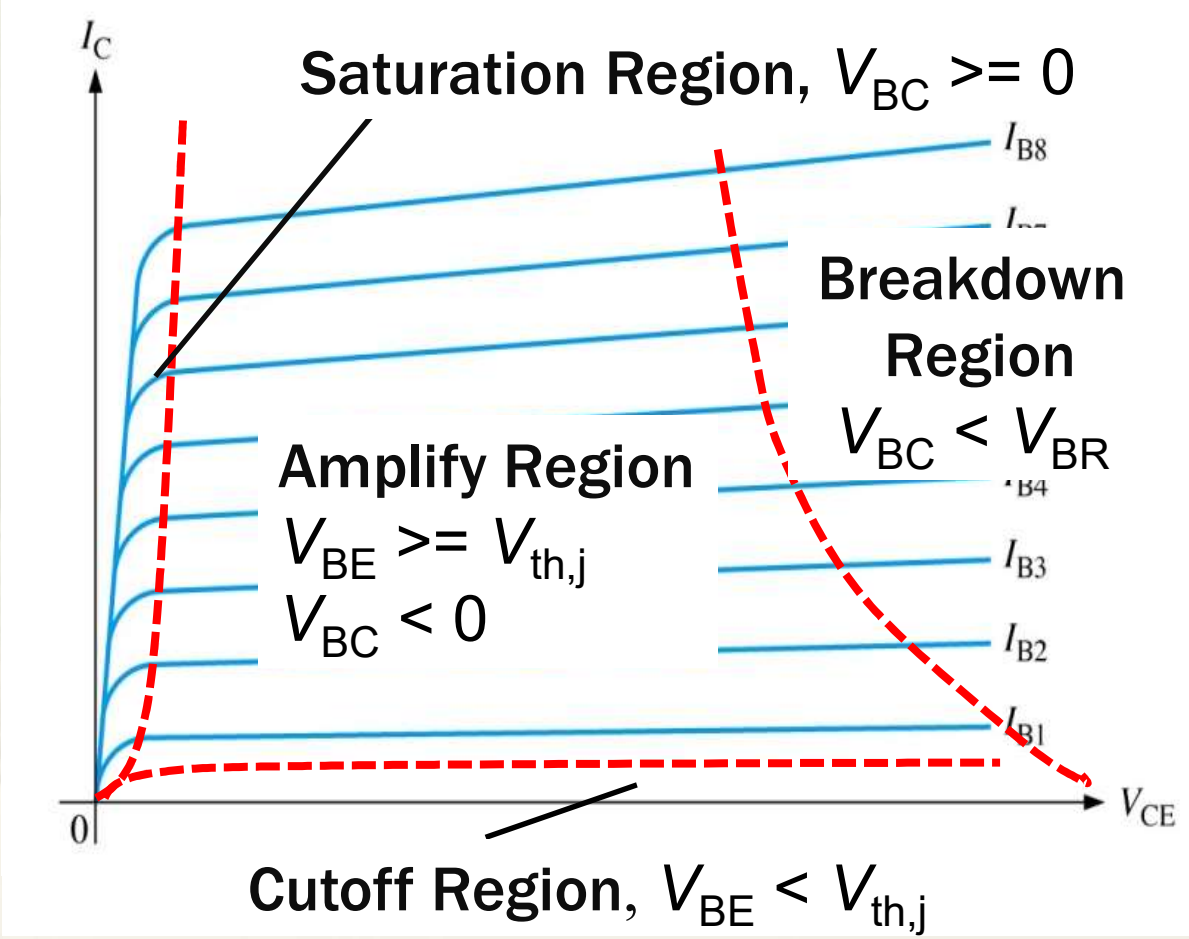


- **Saturation**
  - When the base current is high enough, the entire V_{CC} is dropped across R_C with no voltage (in practice, very small, typically 0.3 ~0.6V) between the collector and emitter. This condition is known as saturation.
  - $I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C} \approx \frac{V_{CC}}{R_C}$

Cutoff and Saturation

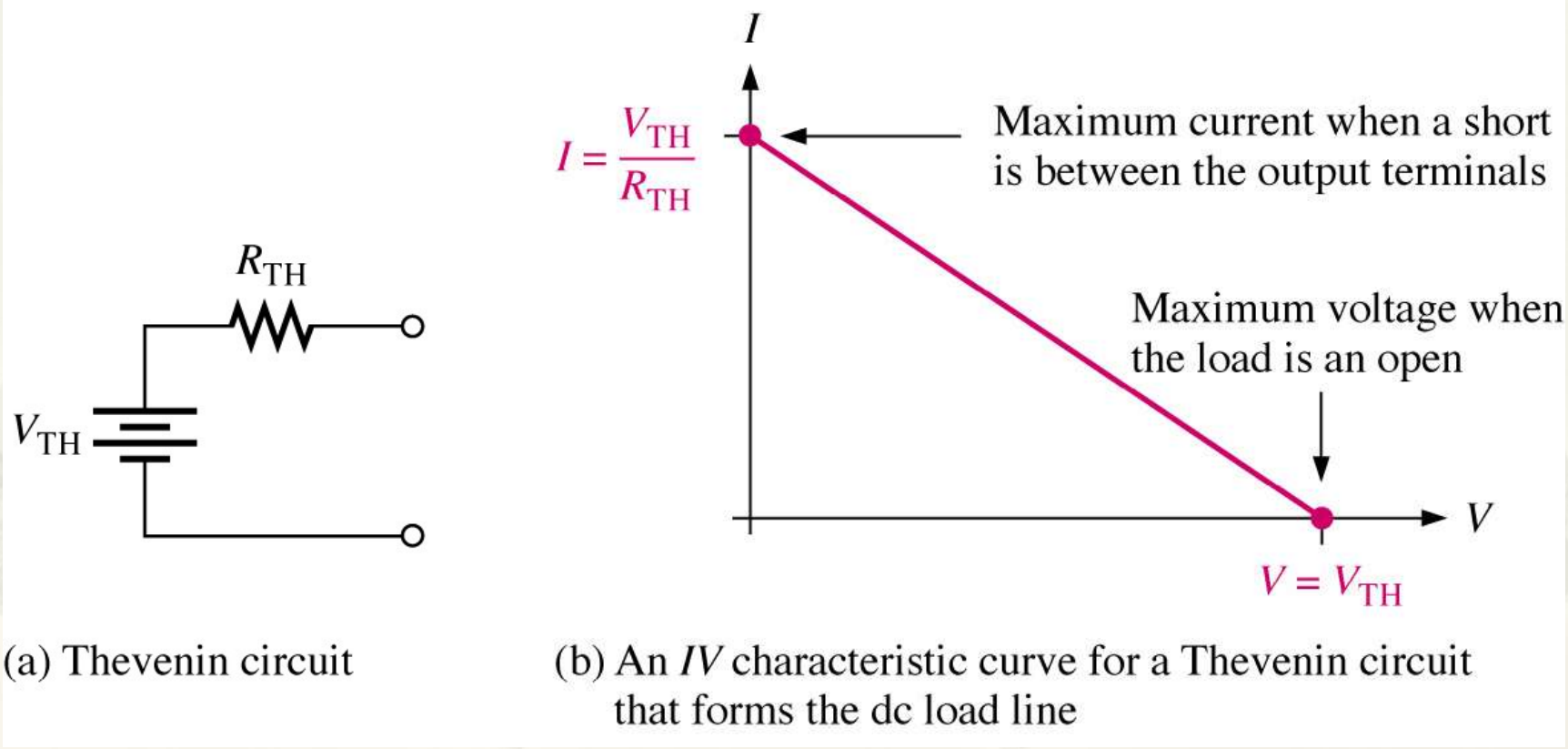
- **Working Regions of a BJT**





DC Load Line

- **Thevenin circuit and IV characteristic curve for the circuit**



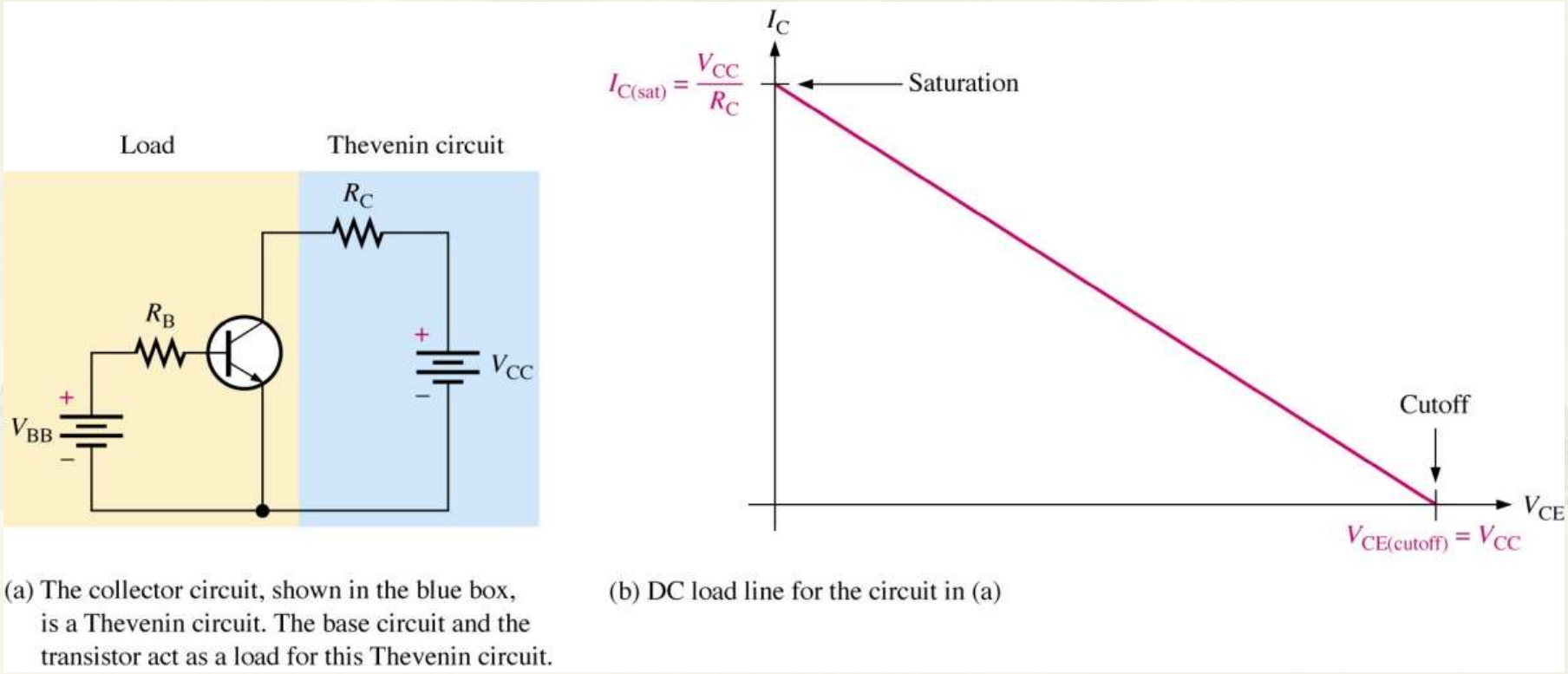
(a) Thevenin circuit

(b) An *IV* characteristic curve for a Thevenin circuit that forms the dc load line

Any load placed across the output terminals will have a current and voltage point located on the load line.

DC Load Line

- A dc load line superimposed on a set of ideal collector characteristic curves



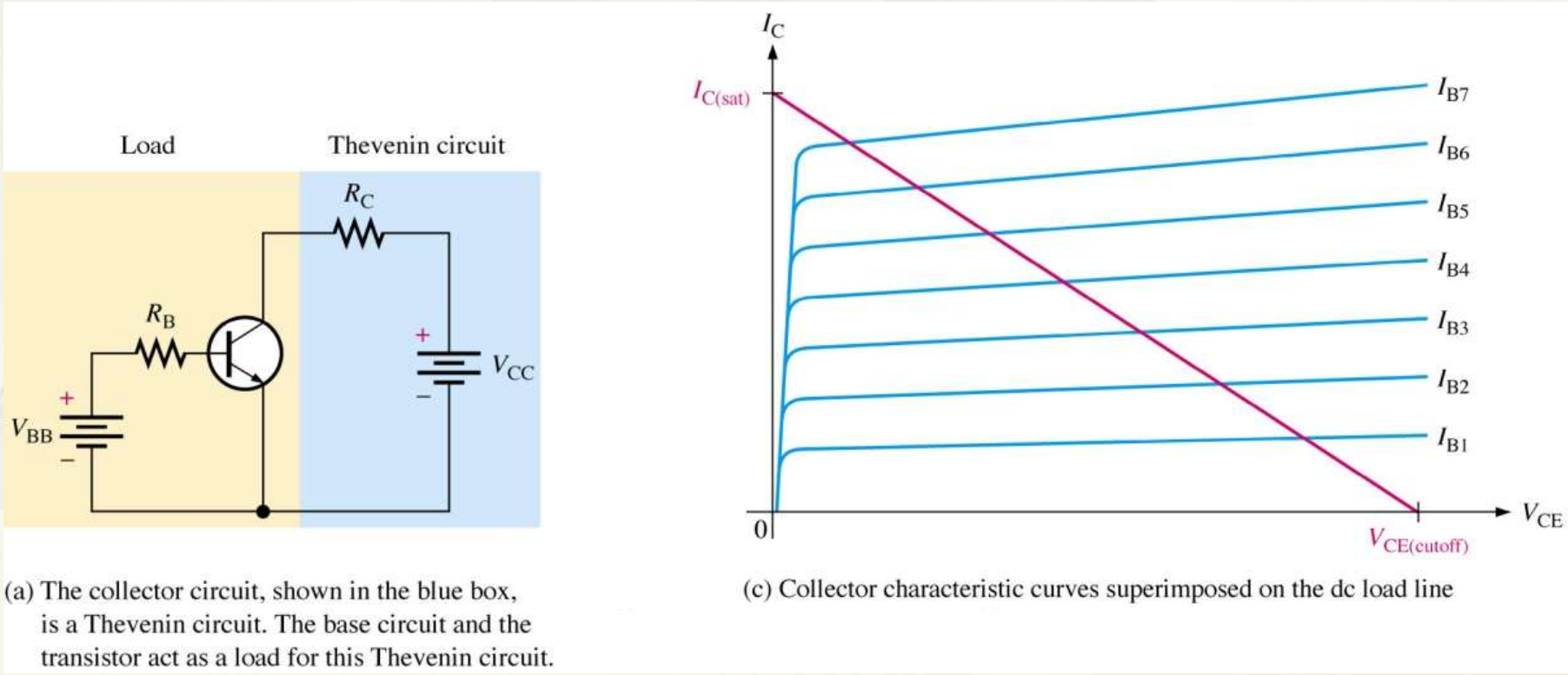
(a) The collector circuit, shown in the blue box, is a Thevenin circuit. The base circuit and the transistor act as a load for this Thevenin circuit.

(b) DC load line for the circuit in (a)

Any value of I_C and the corresponding V_{CE} will fall on this line, as long as dc operation is maintained.

DC Load Line

- A dc load line superimposed on a set of ideal collector characteristic curves

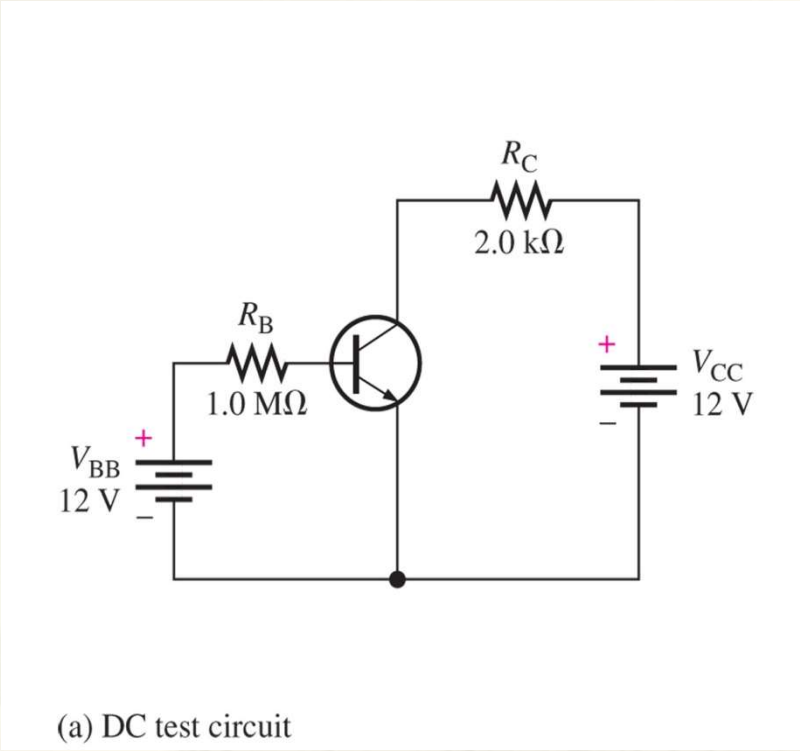


(a) The collector circuit, shown in the blue box, is a Thevenin circuit. The base circuit and the transistor act as a load for this Thevenin circuit.

(c) Collector characteristic curves superimposed on the dc load line

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DC Load Line

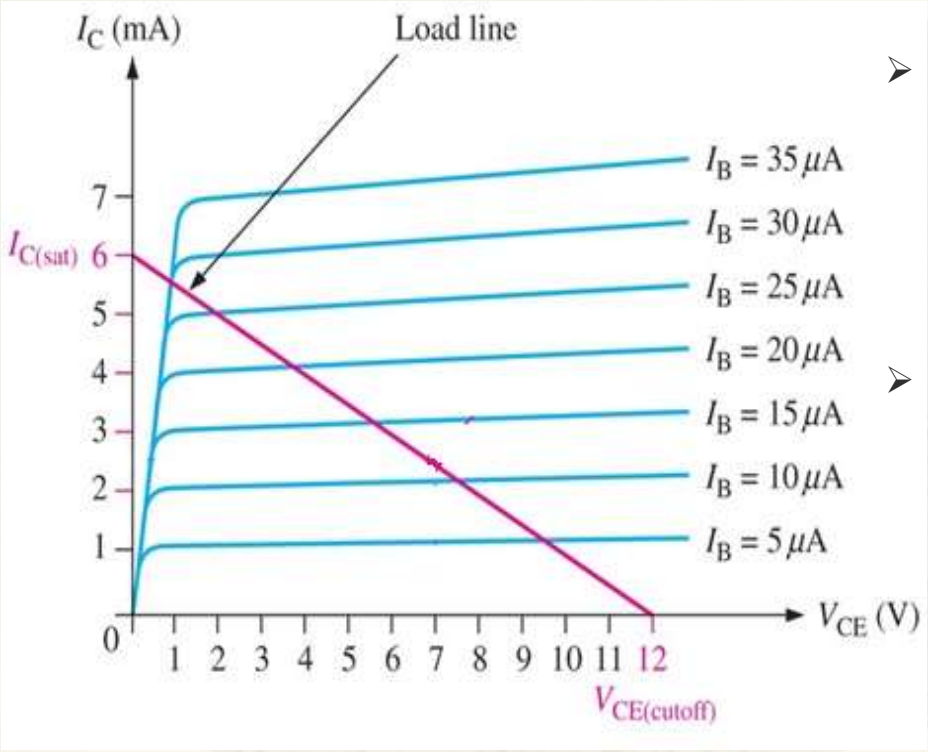


(a) DC test circuit

- **Cutoff**
  - $V_{CE(cutoff)} = V_{CC} = 12V$
  - $I_{C(cutoff)} = 0mA$
- **Saturation**
  - $V_{CE(sat)} = 0V$
  - $I_{CE(sat)} = \frac{V_{CC}}{R_C} = \frac{12V}{2.0k\Omega} = 6.0mA$

The dc load line and the characteristic curves can be combined to illustrate transistor operation

DC Load Line

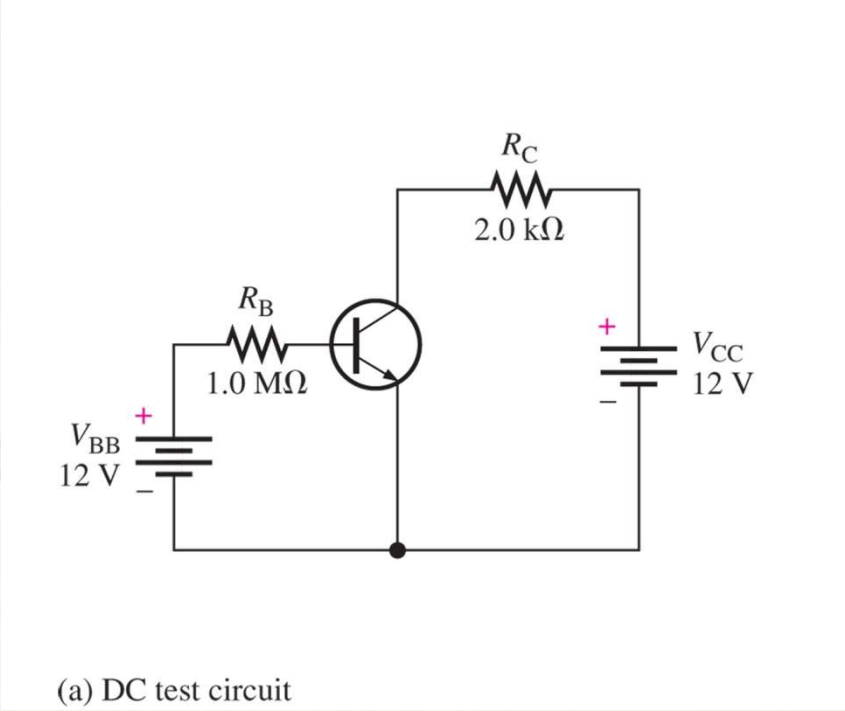


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The dc load line and the characteristic curves can be combined to illustrate transistor operation

Q-Point

- The point at which the actual base current line intersects the load line is the quiescent or Q-point for the circuit.

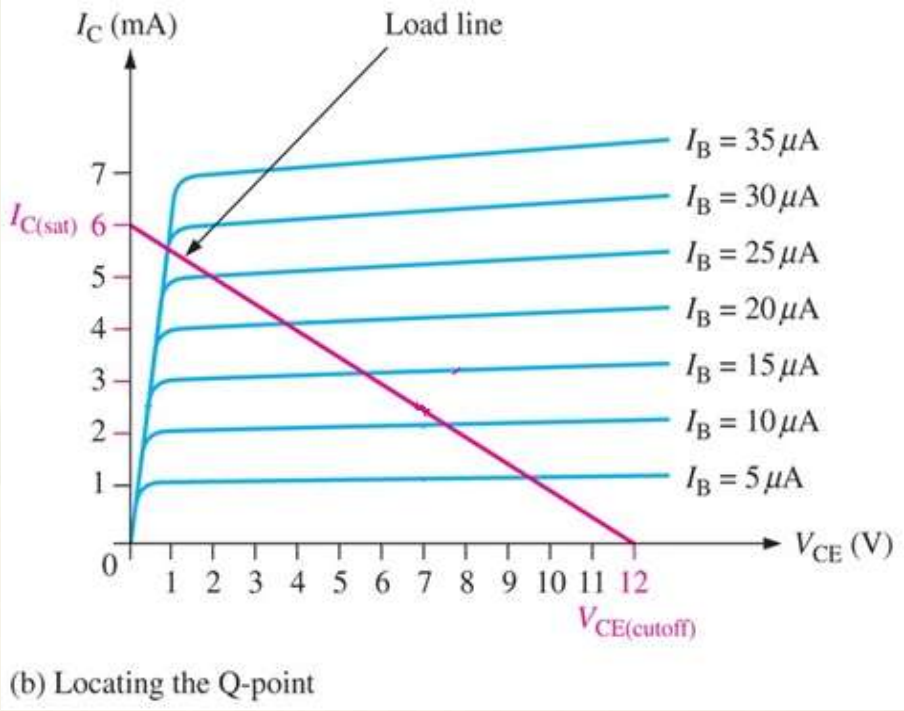


(a) DC test circuit

- Voltage across the base resistor
  - $V_{R_B} = V_{BB} - V_{BE}$
    $= 12\text{V} - 0.7\text{V} = 11.3\text{V}$
- Base current
  - $I_B = \frac{V_{R_B}}{R_B} = \frac{11.3\text{V}}{1.0\text{M}\Omega} = 11.3\mu\text{A}$

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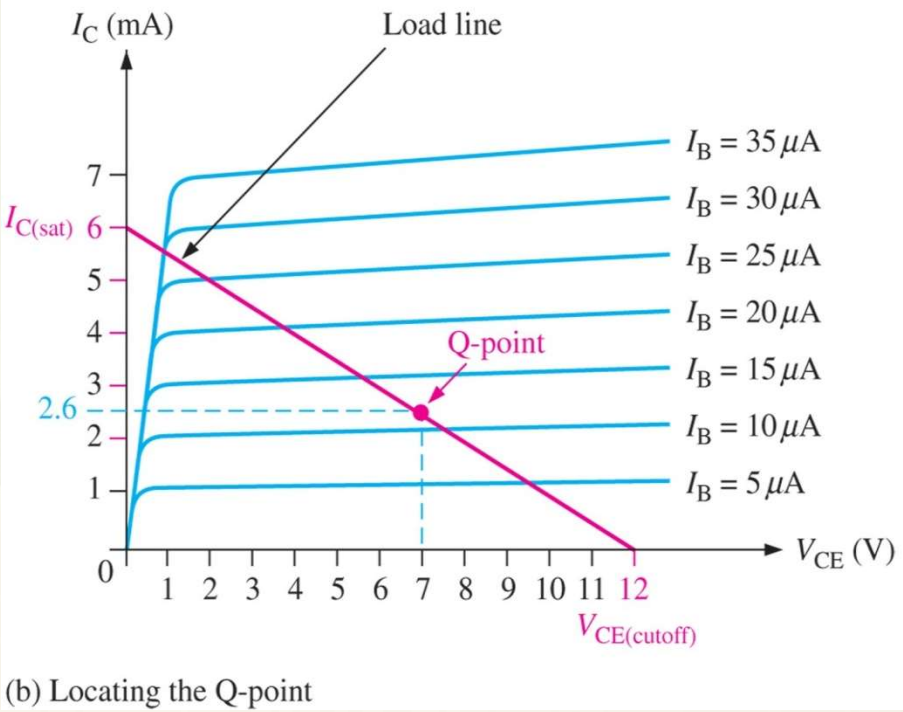


(b) Locating the Q-point

- Voltage across the base resistor
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- Base current
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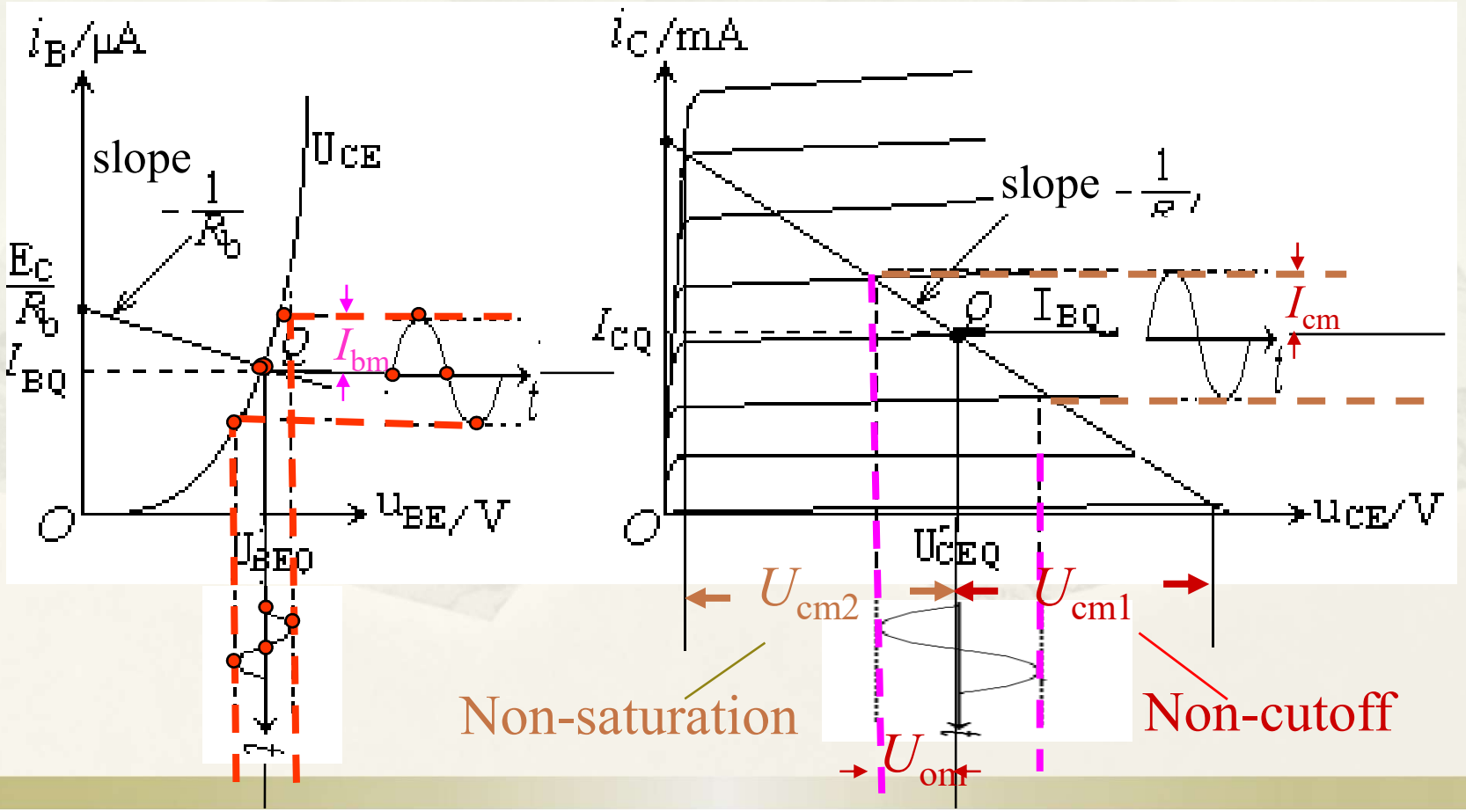


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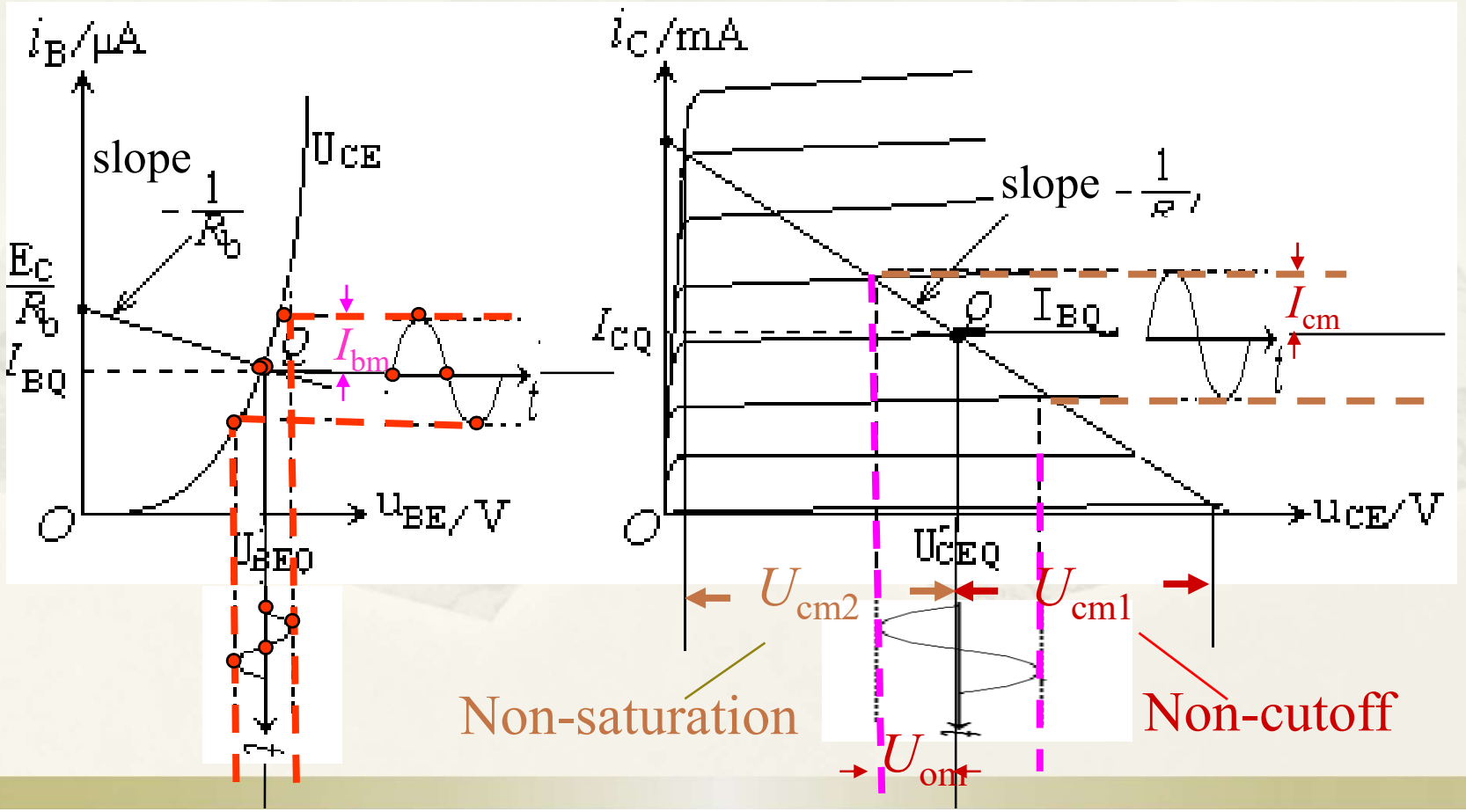
Range of AC Signal amplification

- In order for a transistor to amplify an ac signal, the ac signal needs to be superimposed on a dc level that sets the operating point.



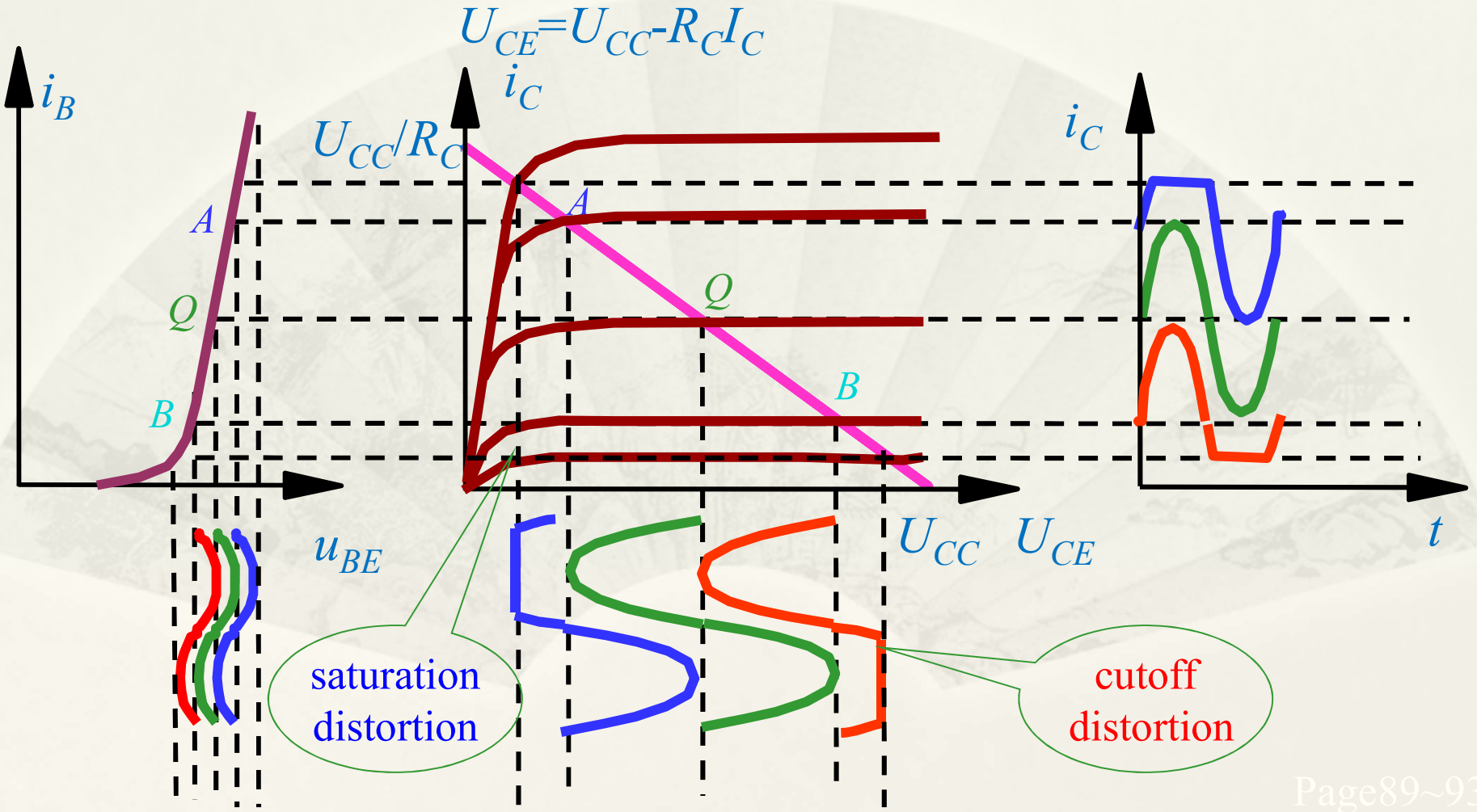
Range of AC Signal amplification

- In order for a transistor to amplify an ac signal, the ac signal needs to be superimposed on a dc level that sets the operating point.



Range of AC Signal amplification

- **Distortion**



BJT Parameters

- **Electrical Characteristics**
  - DC Current Gain(h_{FE})
  - Collector-Emitter Saturation Voltage($V_{CE(sat)}$)
  - Base-Emitter Saturation Voltage($V_{BE(sat)}$)
  - Collector-Emitter Leakage(I_{CEO})
  - Collector-Base Leakage (I_{CBO})
  - Emitter-Base Leakage(I_{EBO})
  - Transition Frequency (f_T)

BJT Parameters

- **Absolute Maximum Ratings**
  - Collector-Emitter Breakdown Voltage $(V_{(\mathbf{BR})\mathbf{CEO}})$
  - Collector-Base Breakdown Voltage $(V_{(\mathbf{BR})\mathbf{CBO}})$
  - Emitter-Base Breakdown Voltage $(V_{(\mathbf{BR})\mathbf{EBO}})$
  - Maximum Collector Current $(I_{\mathbf{CM}})$
  - Maximum Power Dissipation $(P_{\mathbf{CM}})$

Datasheet of an Actual BJT

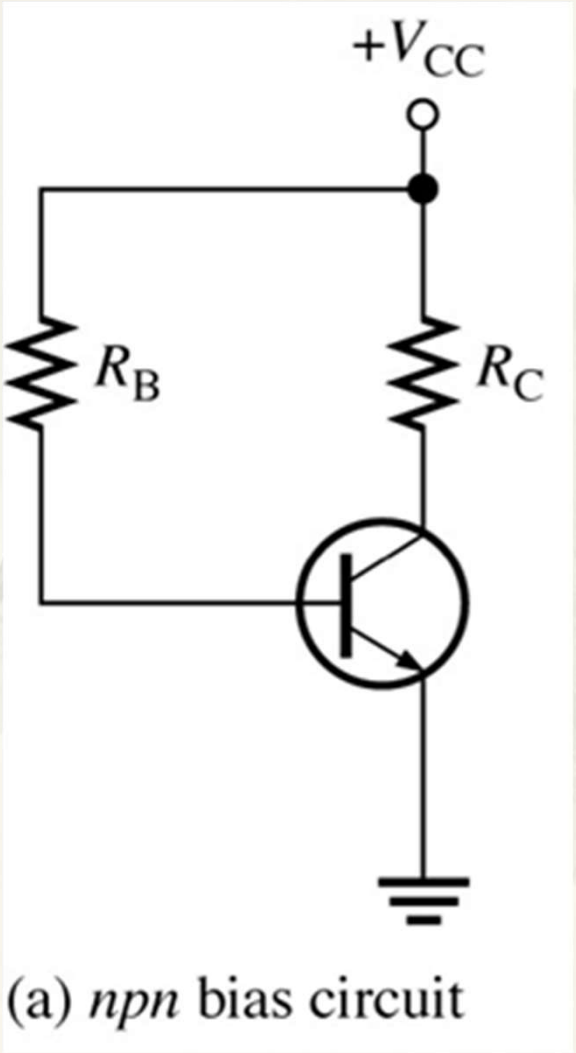
3-2 BJT BIAS CIRCUIT

- Base Bias
- Voltage-Divider Bias
- Emitter Bias
- Troubleshooting A Biased Transistor

Bias Circuit

- **Bias circuit**
  - In order for a transistor to amplify an ac signal, the ac signal needs to be superimposed on a dc level that sets the operating point.
  - Bias circuits set the dc level at a point that allows the ac signal to vary in both the positive and negative directions without driving the transistor into saturation or cutoff.

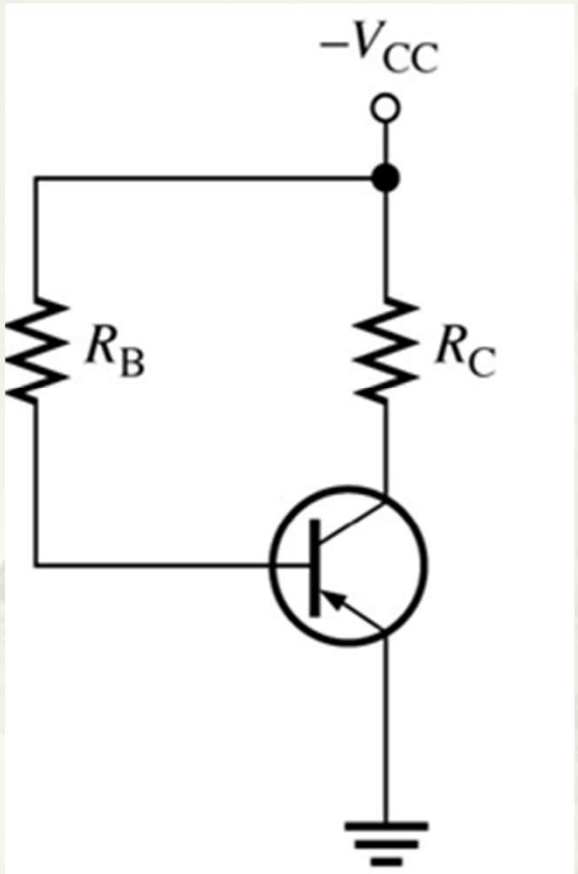
Base Bias



(a) *npn* bias circuit

- Only a single resistor is used for bias
- $I_C = \beta_{DC}\left(\frac{V_{CC}-V_{BE}}{R_B}\right)$
- Only applied to the configuration with no emitter resistor
- Good choice in switching applications

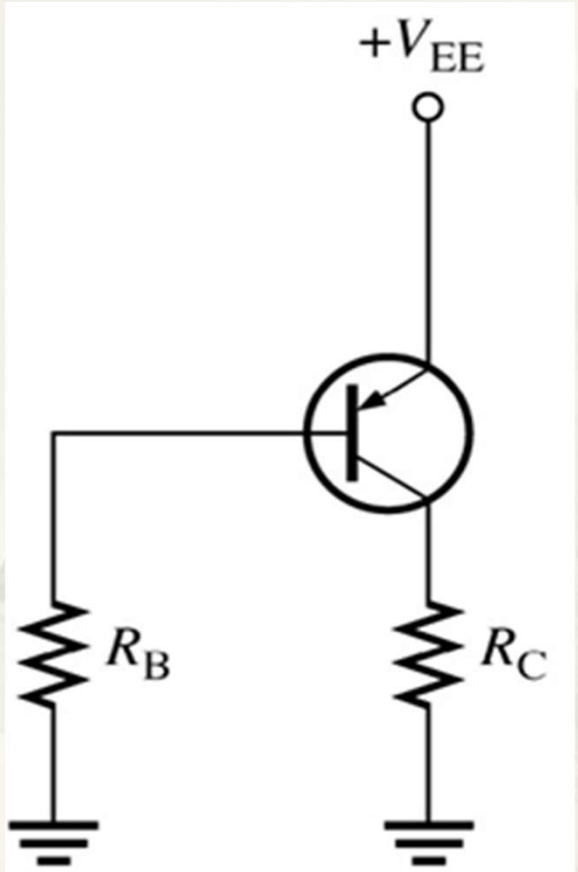
Base Bias



(b) *pnp* bias circuit with a negative supply

- Only a **single resistor** is used for bias
- $I_C = \beta_{DC}\left(\frac{V_{CC}-V_{BE}}{R_B}\right)$
- Only applied to the configuration with **no emitter resistor**
- Good choice in **switching applications**

Base Bias



(c) *pnp* bias circuit with a positive supply

- Only a single resistor is used for bias
- $I_C = \beta_{DC}\left(\frac{V_{CC}-V_{BE}}{R_B}\right)$
- Only applied to the configuration with no emitter resistor
- Good choice in switching applications

Example 3-3

Problem

The manufacturer's specification for a 2N3904 transistor shows that β_{DC} has a range from 100 to 300. Assume a 2N3904 is used in the base-biased circuit shown in Figure 3–16. Compute the minimum and maximum collector current based on this specification. (Note that this is effectively the same circuit that was solved with load line analysis in Figure 3–14 except it now is shown with a single power supply.)

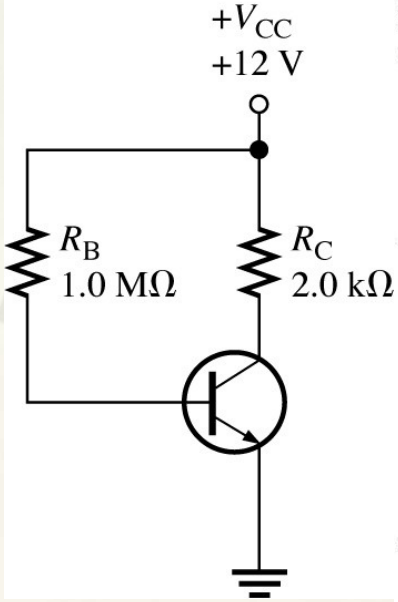


FIGURE 3-16

Solution

The base-emitter junction is forward-biased, causing a 0.7 V drop. The voltage across R_B is

$$V_{R_B} = V_{CC} - V_{BE} = 12\text{ V} - 0.7\text{ V} = 11.3\text{ V}$$

You can find the base current by applying Ohm's law to the base resistor.

$$I_B = \frac{V_{R_B}}{R_B} = \frac{11.3\text{ V}}{1.0\text{ M}\Omega} = 11.3\ \mu\text{A}$$

With linear operation, the collector current is β_{DC} times larger than the base current. Therefore, the minimum collector current is

$$I_{C(min)} = \beta_{DC} I_B = (100)(11.3\ \mu\text{A}) = \mathbf{1.13\ mA}$$

Example 3-3

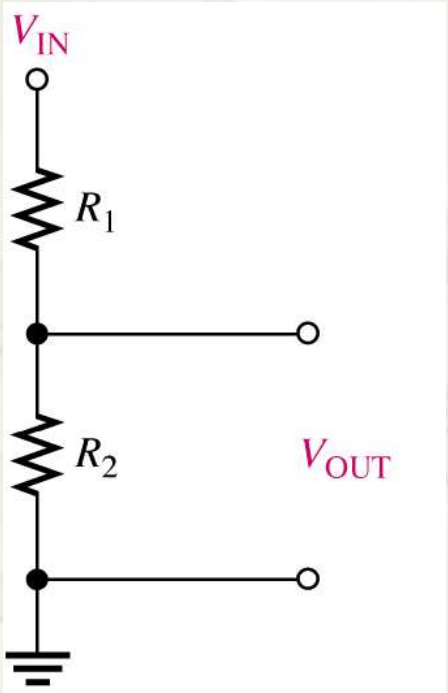
Solution

The maximum collector current is

$$I_{C(max)} = \beta_{DC} I_B = (300)(11.3\ \mu A) = \mathbf{3.39\ mA}$$

Voltage-Divider Bias

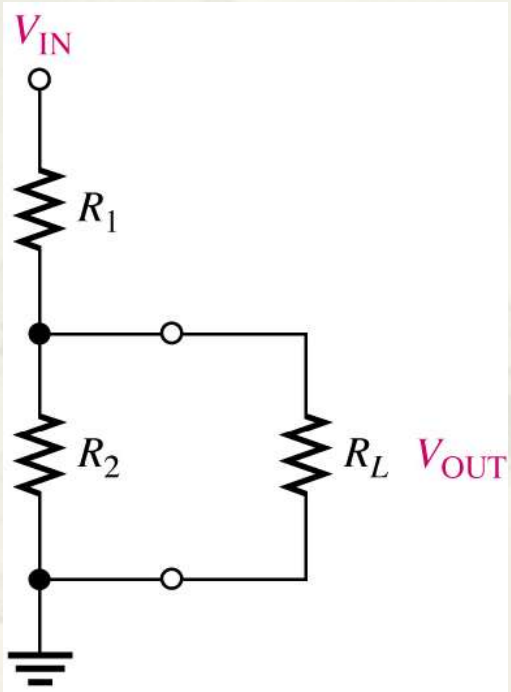
- **Disadvantage of base bias**
  - Dependency on β_{DC}
- **Voltage divider**
  - $V_{OUT} = \left(\frac{R_2}{R_1+R_2}\right) V_{IN}$



(a) Unloaded

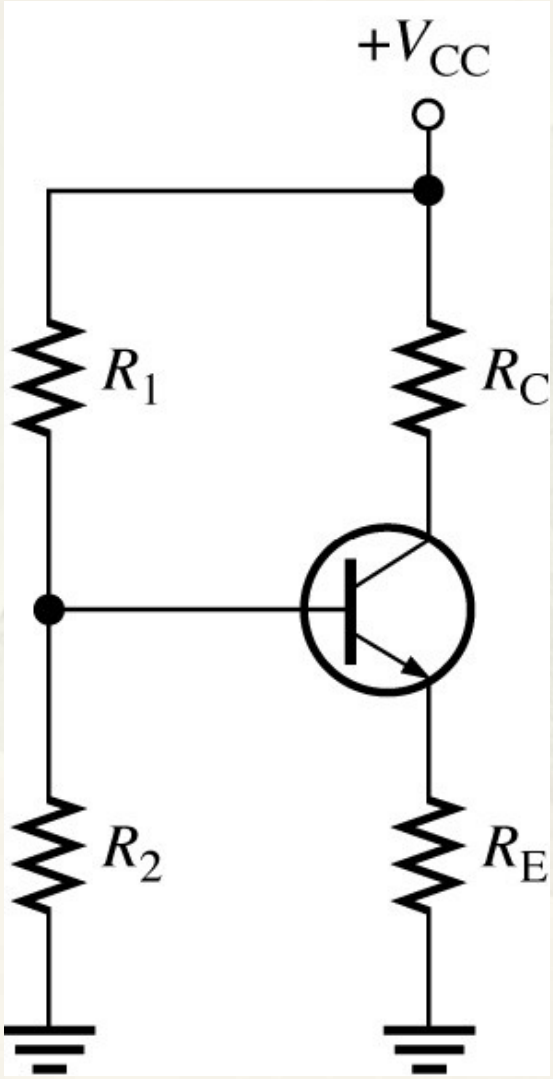
Voltage-Divider Bias

- **Disadvantage of base bias**
  - Dependency on β_{DC}
- **Voltage divider**
  - $V_{OUT} = \left(\frac{R_2}{R_1+R_2}\right) V_{IN}$
- **Loading effect**
  - Loading effect is a change in circuit parameters when a load is connected.



(b) Loaded

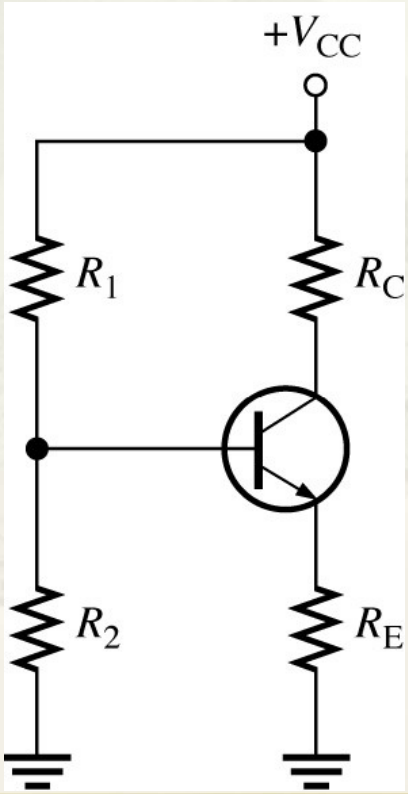
Voltage-Divider Bias



- With voltage-divider bias (stiff bias), the transistor acts as a high resistance load on the divider.
- As a rule of thumb, these resistor should have a current that is at least 10 times larger than the base current to avoid variation in the base voltage when a transistor with a different β_{DC} is used.

Voltage-Divider Bias

- **The steps in computing the parameters for a voltage-divider bias circuit**
  - Based on the assumption of no significant loading effect
  - $V_B = \left(\frac{R_2}{R_1 + R_2}\right) V_{CC}$
  - $V_E = V_B - 0.7\text{V}$
  - $I_E = \frac{V_E}{R_E}, I_C \cong I_E$
  - $V_C = V_{CC} - I_C R_C$
  - $V_{CE} = V_C - V_E$



Example 3-4

Problem

Find V_B, V_E, I_E, I_C, and V_{CE} for the circuit in Figure 3–19.

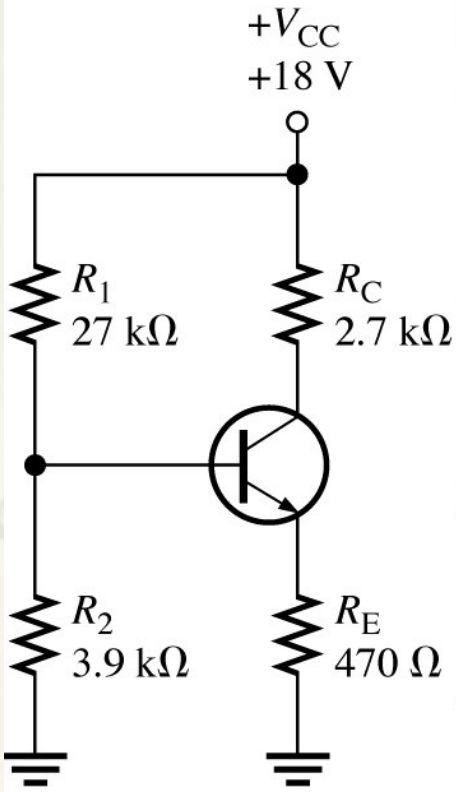


FIGURE 3-19

Solution

Begin by finding the base voltage using the voltage-divider rule.

$$V_B = \left(\frac{R_2}{R_1 + R_2}\right)V_{CC} = \left(\frac{3.9\ \text{k}\Omega}{27\ \text{k}\Omega + 3.9\ \text{k}\Omega}\right)18\ \text{V} = \mathbf{2.27\ V}$$

The emitter voltage is one diode drop less than the base voltage.

$$V_E = V_B - V_{BE} = 2.27\ \text{V} - 0.7\ \text{V} = \mathbf{1.57\ V}$$

Next, find the emitter current from Ohm's law.

$$I_E = \frac{V_E}{R_E} = \frac{1.57\ \text{V}}{470\ \Omega} = \mathbf{3.34\ mA}$$

Using the approximation $I_C \cong I_E$,

$$I_C = \mathbf{3.34\ mA}$$

Example 3-4

Now find the collector voltage.

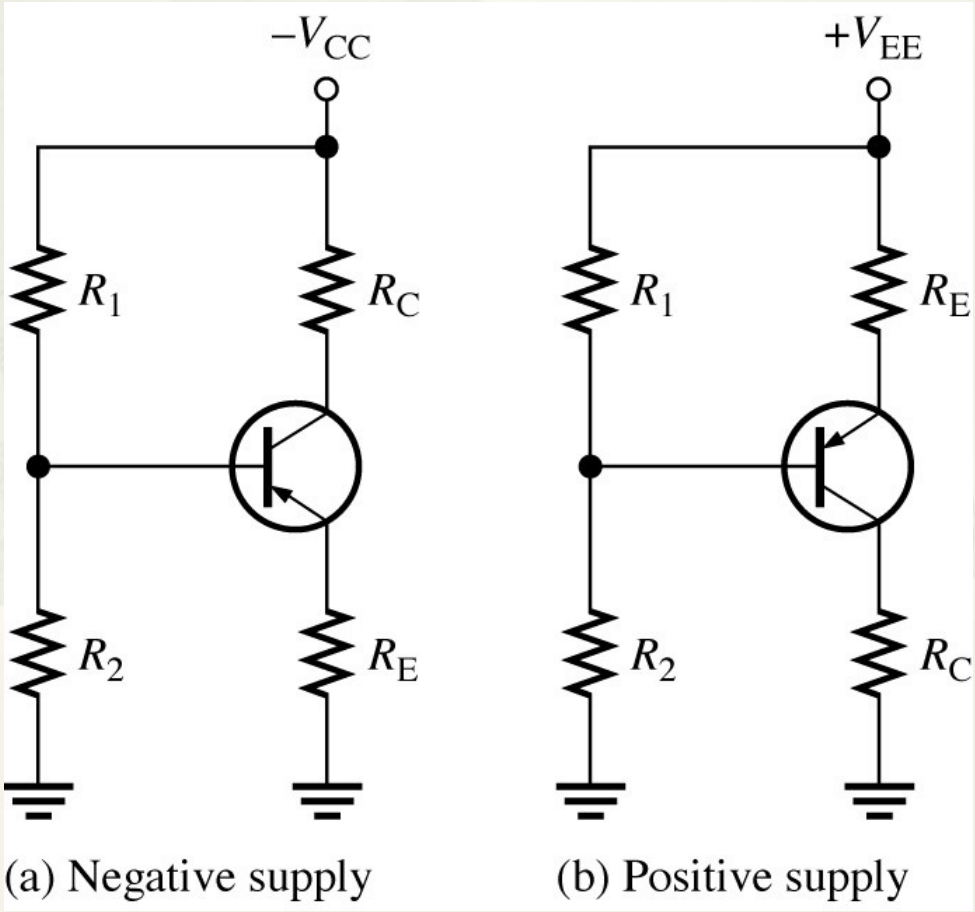
$$V_C = V_{CC} - I_C R_C = 18\ \text{V} - (3.34\ \text{mA})(2.7\ \text{k}\Omega) = 8.98\ \text{V}$$

The collector-emitter voltage is

$$V_{CE} = V_C - V_E = 8.98\ \text{V} - 1.57\ \text{V} = \mathbf{7.41\ V}$$

Voltage-Divider Bias

- **Voltage-divider bias with a *pnp* transistor**
  - The equations for *npn* transistor can be applied to *pnp* transistors, but you need to be careful of algebraic signs.



Example 3-5

Problem

Find V_B, V_E, I_E, I_C, and V_{CE} for the *pnp* circuit in Figure 3–21.

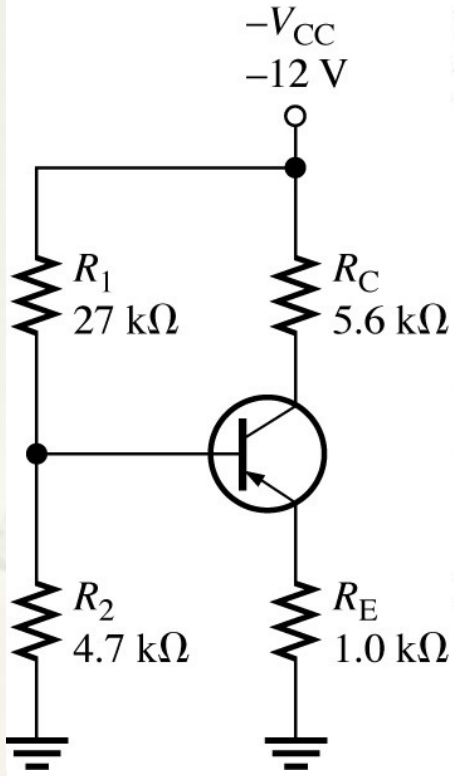


FIGURE 3-21

Solution

Begin by finding the base voltage using the voltage-divider rule.

$$V_B = \left(\frac{R_2}{R_1 + R_2}\right)V_{CC} = \left(\frac{4.7\ \text{k}\Omega}{27\ \text{k}\Omega + 4.7\ \text{k}\Omega}\right)(-12\ \text{V}) = \mathbf{-1.78\ V}$$

The equation for V_E is the same one used for the *npn* transistor but note the signs. The emitter voltage is one diode drop *greater* than the base voltage for a forward-biased *pnp* transistor.

$$V_E = V_B - V_{BE} = -1.78 - (-0.7\ \text{V}) = \mathbf{-1.08\ V}$$

Now find the emitter current using Ohm's law.

$$I_E = \frac{V_E}{R_E} = \frac{-1.08\ \text{V}}{1.0\ \text{k}\Omega} = \mathbf{-1.08\ mA}$$

Using the approximation $I_C \cong I_E$,

$$I_C = \mathbf{-1.08\ mA}$$

Example 3-5

Now find the collector voltage.

$$V_C = V_{CC} - I_C R_C = -12\text{ V} - (-1.08\text{ mA})(5.6\text{ k}\Omega) = -5.96\text{ V}$$

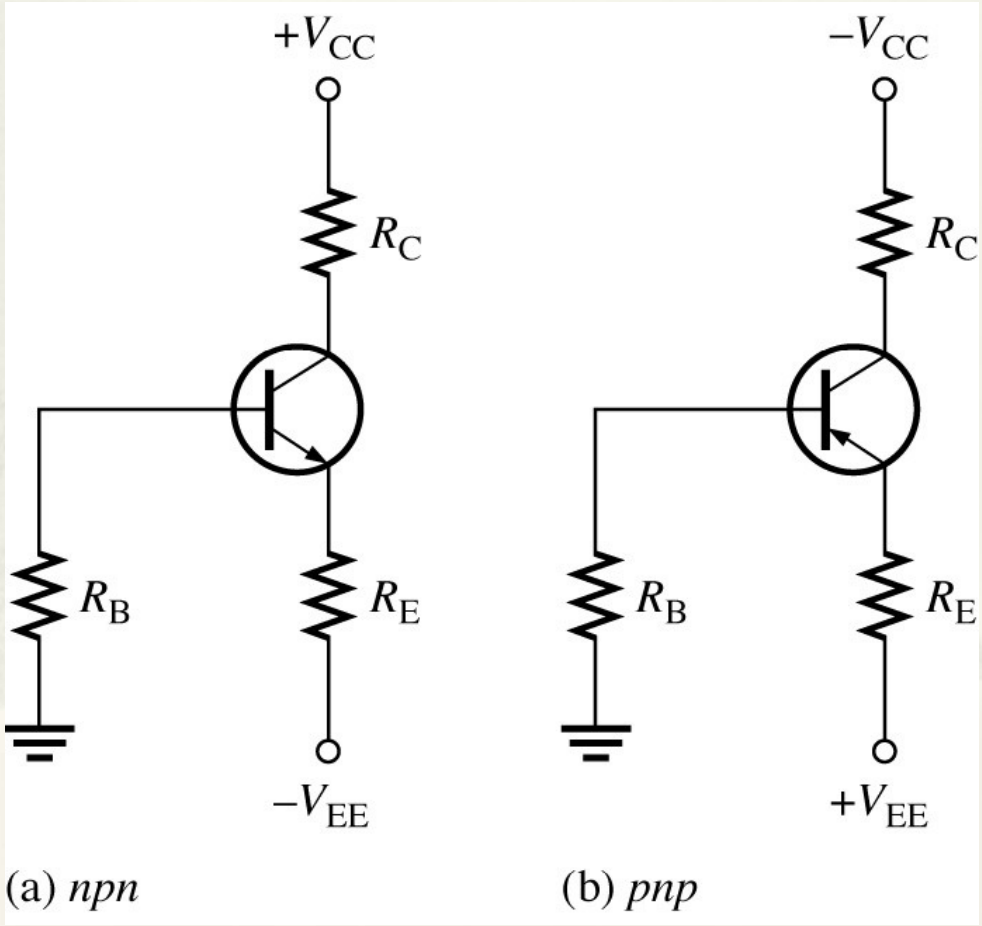
The collector-emitter voltage is

$$V_{CE} = V_C - V_E = -5.96\text{ V} - (-1.08\text{ V}) = \mathbf{-4.88\text{ V}}$$

Notice that V_{CE} is negative for a *pnp* circuit.

Emitter Bias

- **Emitter Bias**
  - Very stable
  - Used in most integrated circuit amplifiers
  - Disadvantage of requiring two power supplies
  - $I_C = \frac{V_{R_E}}{R_E}$
  - $V_C = V_{CC} - I_C R_C$



Example 3-6

Problem

Find V_E, I_E, I_C, and V_{CE} for the emitter bias circuit in Figure 3–23.

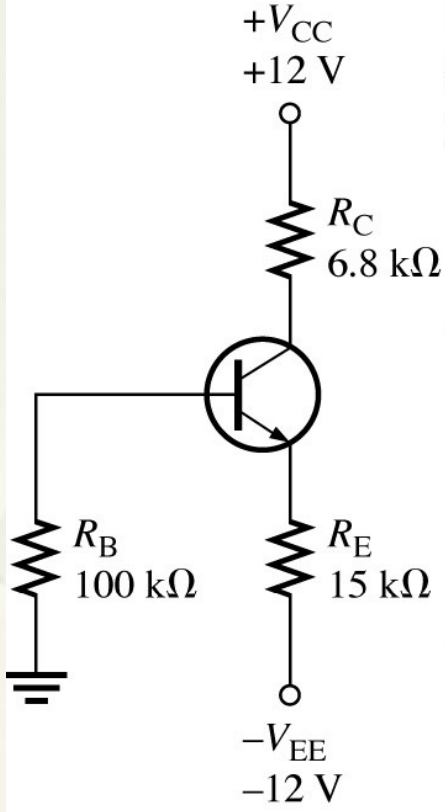


FIGURE 3-23

Solution

Start with the approximation $V_E \cong \mathbf{-1\ V}$. This implies the voltage across R_E is 11 V. Applying Ohm's law to the emitter resistor,

$$I_E = \frac{V_{R_E}}{R_E} = \frac{11\ \text{V}}{15\ \text{k}\Omega} = \mathbf{0.73\ mA}$$

The collector current is approximately equal to the emitter current.

$$I_C \cong \mathbf{0.73\ mA}$$

Now find the collector voltage.

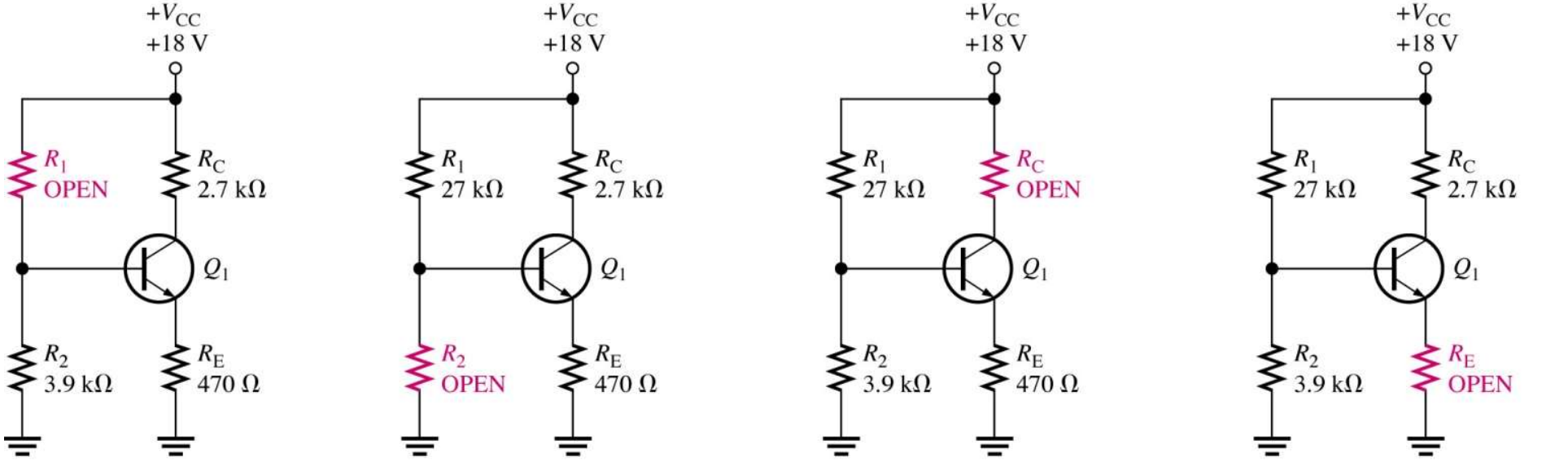
$$V_C = V_{CC} - I_C R_C = 12\ \text{V} - (0.73\ \text{mA})(6.8\ \text{k}\Omega) = 7.0\ \text{V}$$

Find V_{CE} by subtracting V_E from V_C.

$$V_{CE} = V_C - V_E = 7.0\ \text{V} - (-1\ \text{V}) = \mathbf{8.0\ V}$$

Troubleshooting A Biased Transistor

- **Typical open circuit problems in a voltage-divider biased transistor**



(a) *Fault:* open R_1
Symptoms: transistor cut off
$V_B = 0$ V
$V_E = 0$ V
$V_C = 18$ V

(b) *Fault:* open R_2
Symptoms: transistor saturated
$V_B = +18$ V
$V_E = 17.3$ V
$V_C = 17.3$ V

(c) *Fault:* open R_C
Symptoms: transistor cut off
$V_B = 0.9$ V
$V_E = 0.2$ V (due to less current)
$V_C = 0.2$ V

(d) *Fault:* open R_E
Symptoms: transistor cut off
$V_B = 2.3$ V
$V_E = 1.9$ V
$V_C = 18$ V

3-3 AMPLIFIER CONCEPTS AND AC CONSIDERATIONS

- Linear Amplifiers
- Nonlinear Amplifiers
- DC and AC Quantities
- AC Equivalent Circuits

Linear Amplifiers

- **Linear amplifier**
  - Produces a magnified replica of the input signal in order to produce a useful outcome
- **Ideal linear amplifier**
  - The amplifier introduces no noise or distortion to the signal
  - The output varies in time and replicates the input exactly
- **Gain**
  - The amount of amplification is called gain
- **Voltage gain**
  - The ratio of output voltage to input voltage
  - $A_v = \frac{V_{out}}{V_{in}}$

Example 3-7

Problem

The input and output signals for a linear amplifier are shown in Figure 3–26 and represent an oscilloscope display. What is the voltage gain of the amplifier?

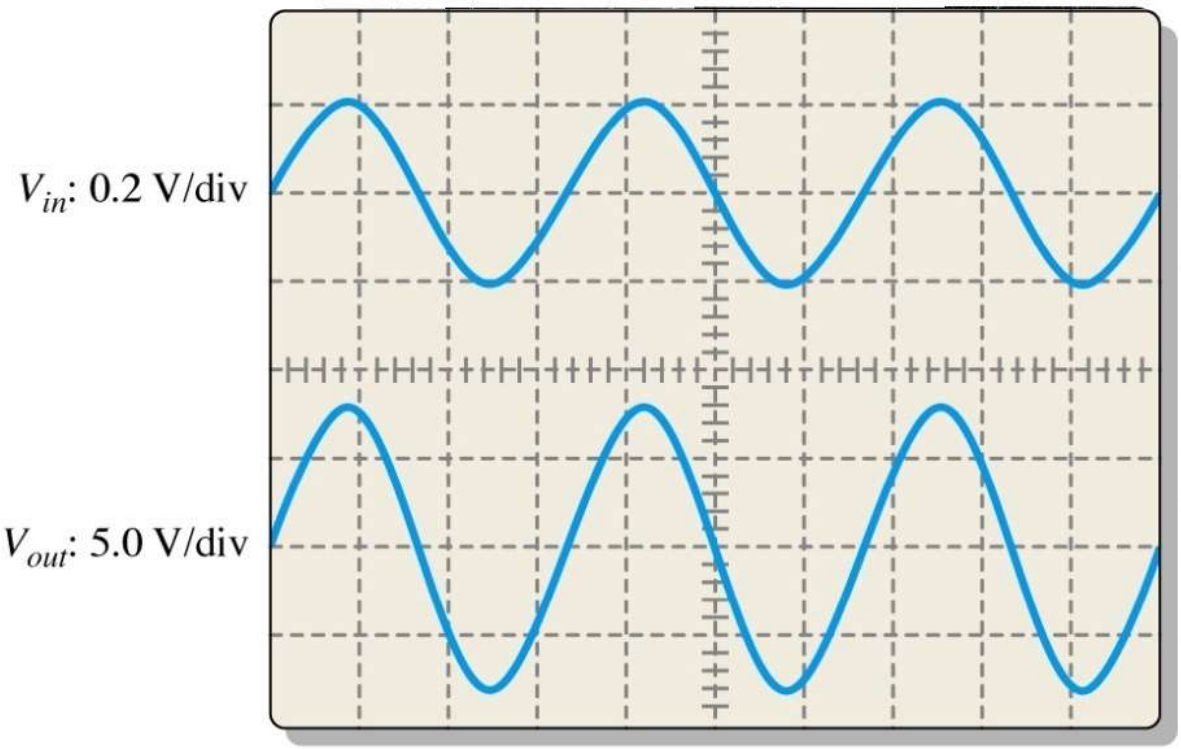


FIGURE 3–26
Oscilloscope display.

Example 3-7

Solution

The input signal is 2.0 divisions from peak-to peak.

$$V_{in} = 2.0 \text{ div} \times 0.2 \text{ V/div} = 0.4 \text{ V}$$

The output signal is 3.2 divisions from peak-to peak.

$$V_{out} = 3.2 \text{ div} \times 5.0 \text{ V/div} = 16 \text{ V}$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{16 \text{ V}}{0.4 \text{ V}} = \mathbf{40}$$

Note that voltage gain is a ratio of voltages and therefore has no units. The answer is the same if rms or peak values had been used for both the input and output voltages.

Nonlinear Amplifier

- **Nonlinear amplifier**
  - Waveshaping amplifier
  - Switching amplifier
- **Power gain (A_p)**
  - The ratio of the signal power out to the signal power in
  - $$A_p = \frac{P_{out}}{P_{in}} = \frac{I_{out}V_{out}}{I_{in}V_{in}} = \left(\frac{V^2_{out}/R_L}{V^2_{in}/R_{in}}\right)$$

No amplifier creates power; it can only convert some of the power from a dc source into signal power.

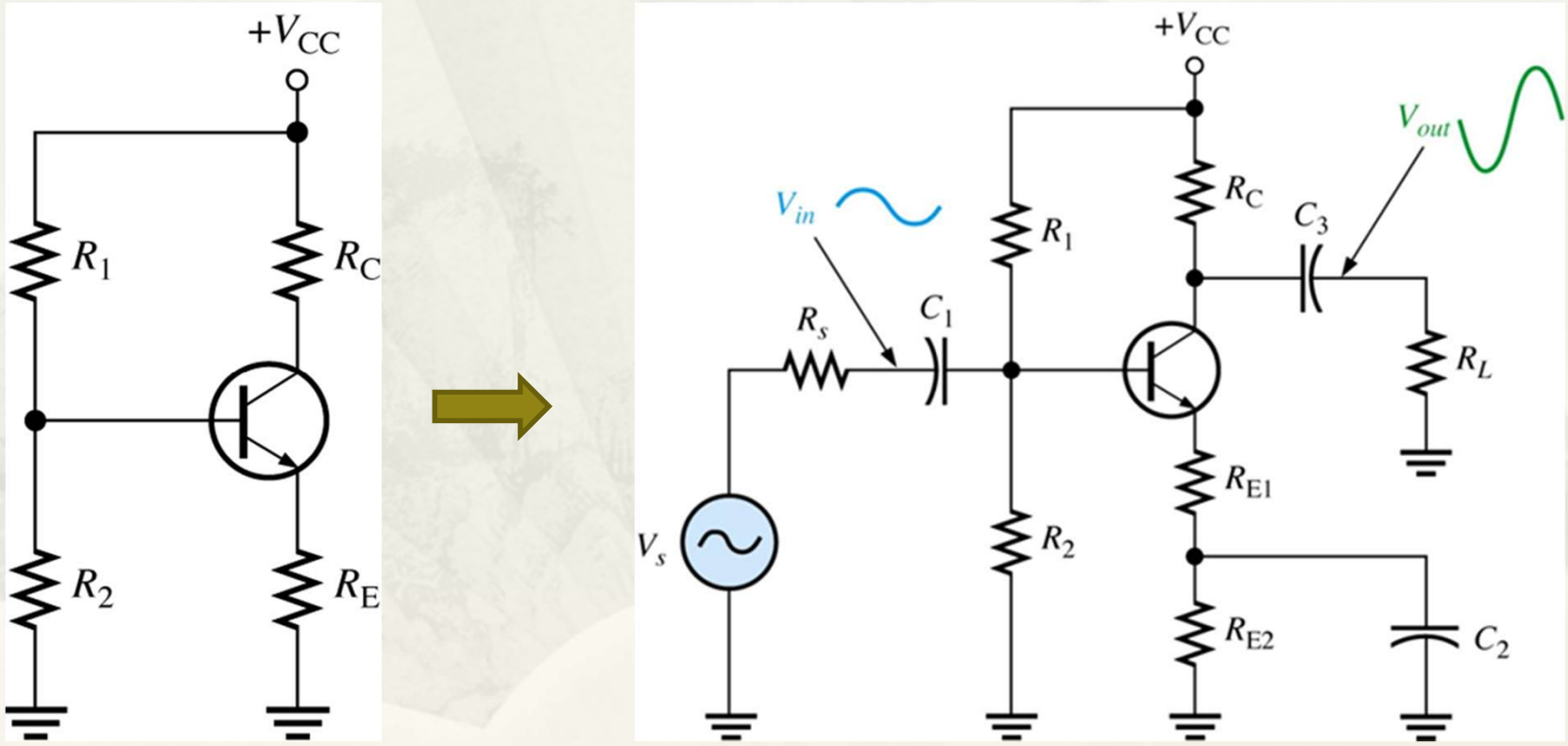
DC and AC Quantities

- **DC quantities, DC resistances**
  - Standard italic capital letters with nonitalic capital-letter subscripts
  - $V_{\mathrm{E}}, I_{\mathrm{E}}, I_{\mathrm{C}}, V_{\mathrm{CE}}, R_{\mathrm{C}}$
- **AC quantities, AC resistances**
  - Standard italic capital letters with lowercase italic subscripts
  - $V_e, I_e, I_c, V_{ce}, R_c$
- **Instantaneous quantities, internal resistances**
  - Both lowercase italic letters and subscripts
  - $v_e, i_e, i_c, v_{ce}, r'_e$
- **The total of DC and AC**
  - Lowercase italic letters with nonitalic capital-letter subscripts
  - $v_{\mathrm{E}}, i_{\mathrm{B}}, i_{\mathrm{C}}, v_{\mathrm{CE}}$

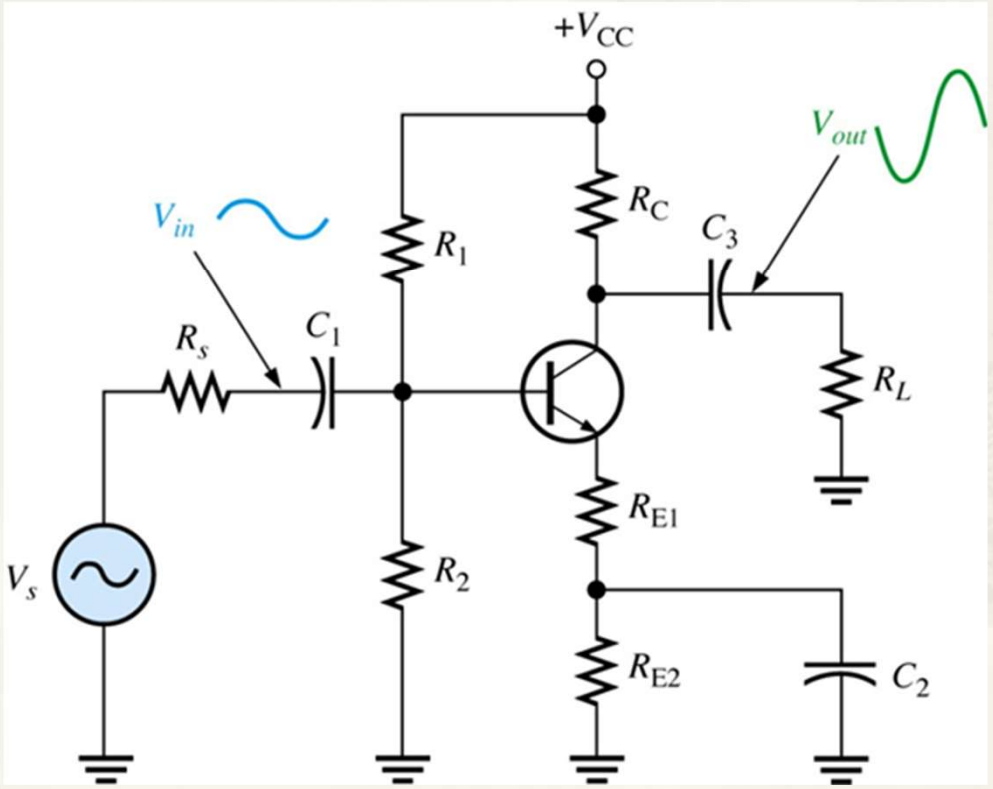
AC Equivalent Circuit

- **Steps in analyzing or troubleshooting a transistor circuit**
  1. Find the dc conditions.
  2. Check ac signals.
- **DC equivalent circuits**
  - Mentally replacing capacitor with an open.
- **AC equivalent circuits**
  - Mentally replacing power supply and capacitor with a short.
- **AC ground**
  - V_{CC} is actually at ground potential for the ac signal.

AC Equivalent Circuit

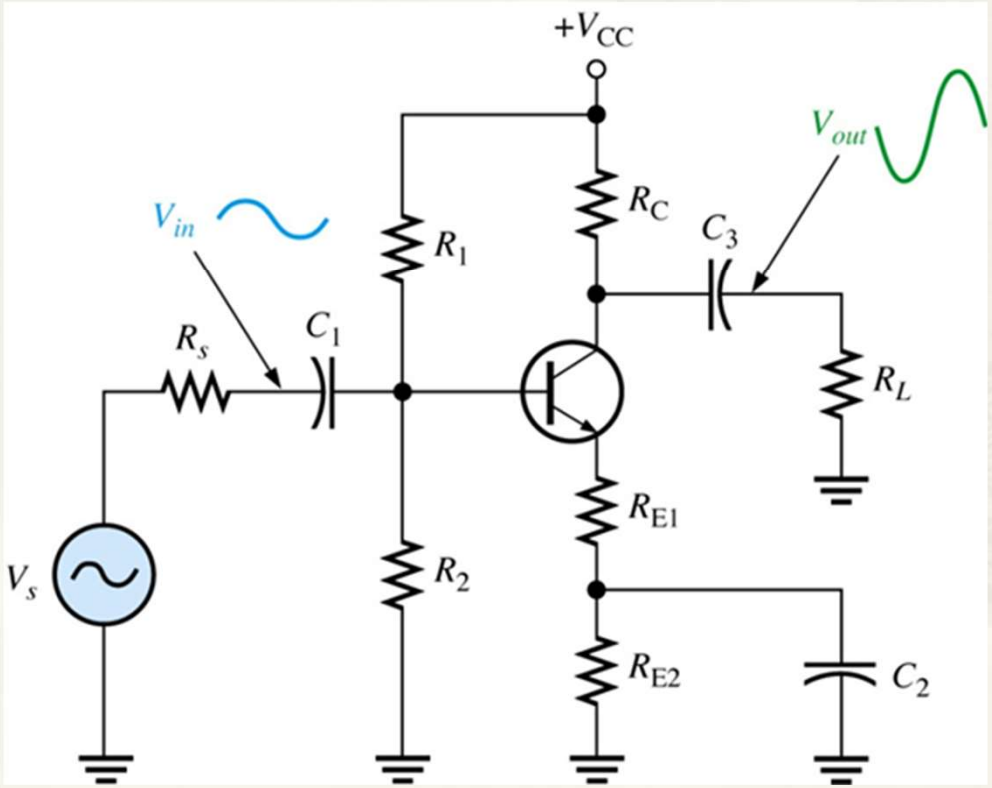


AC Equivalent Circuit



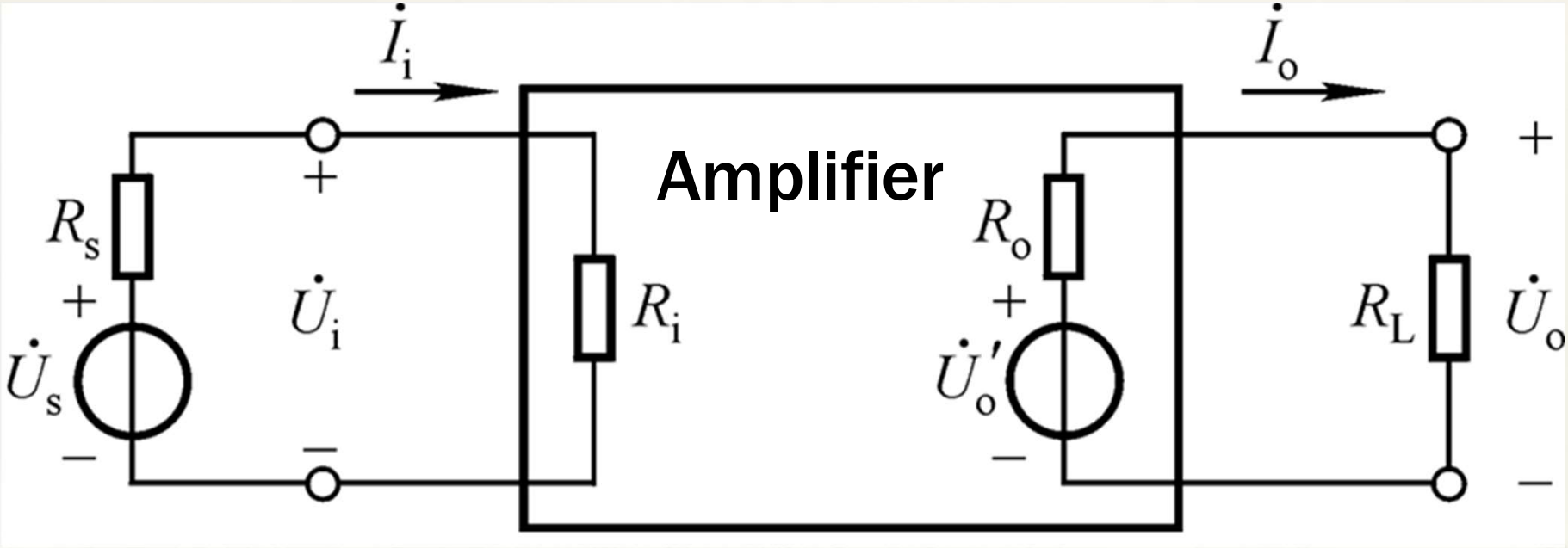
- **Coupling Capacitors**
  - Connected in series with the signal path(C_1, C_3)
  - To pass the ac signal while simultaneously block the dc voltage
- **Bypassing Capacitors**
  - Connected in parallel with an emitter resistor(C_2)
  - To increase the gain of the amplifier

AC Equivalent Circuit



- **Amplification**
  - The sinusoidal collector-to-emitter voltage varies above and bellow its Q-point value 180° out of phase with the base voltage
  - A small swing in base current produces a large variation in collector voltage

AC Equivalent Circuit



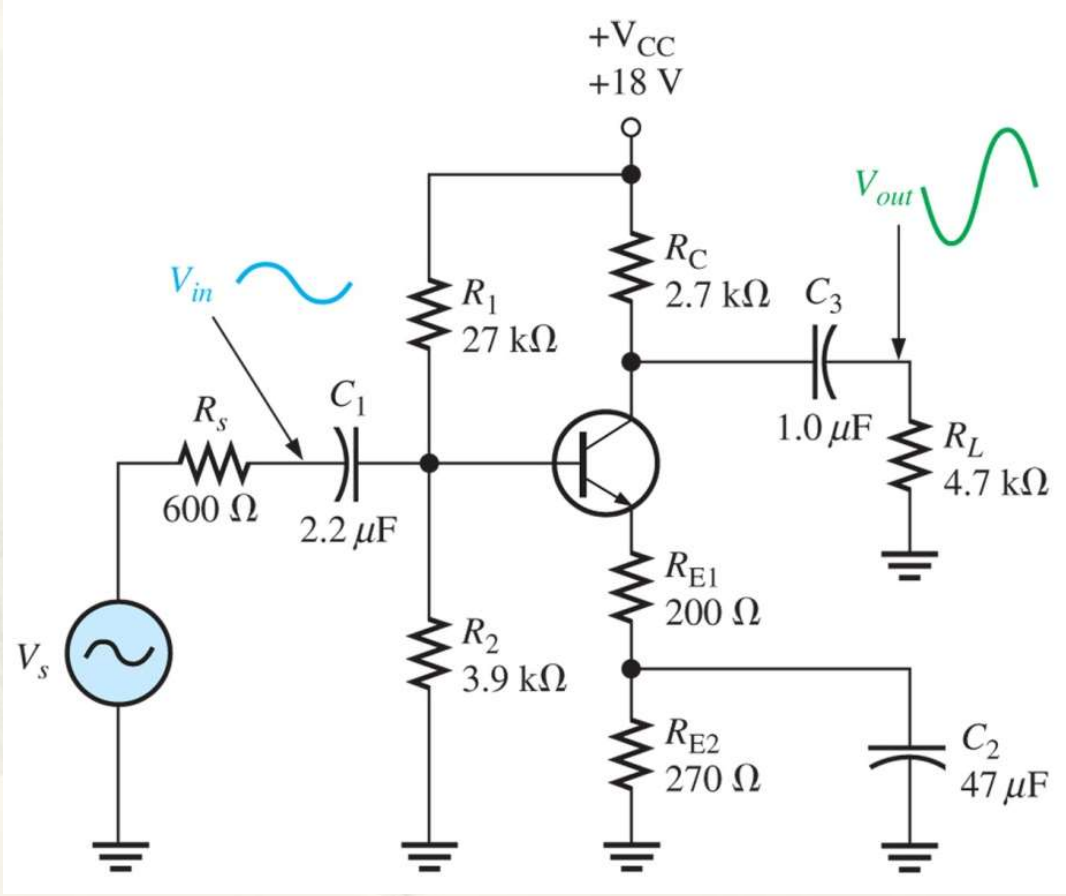
- **Input Resistance (Impedance)**
  - The larger the better (at low frequency)
- **Output Resistance (Impedance)**
  - The smaller the better (at low frequency)

3-4 SMALL-SIGNAL AMPLIFIERS

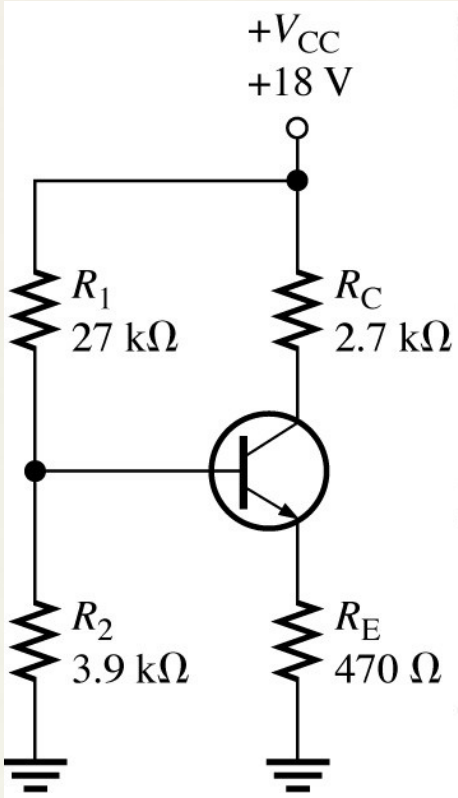
- Common-Emitter Amplifier
- Common-Collector Amplifier
- Common-Base Amplifier

The Common-Emitter Amplifier

- **Common-emitter (CE) amplifier**
  - The most widely used type of BJT amplifier
  - Which has the emitter as the reference terminal for the input and output signal



Whenever you calculate parameters for a transistor amplifier, always start with the dc parameters.



Solution

Begin by finding the base voltage using the voltage-divider rule.

$$V_B = \left(\frac{R_2}{R_1 + R_2}\right)V_{CC} = \left(\frac{3.9\ \text{k}\Omega}{27\ \text{k}\Omega + 3.9\ \text{k}\Omega}\right)18\ \text{V} = \mathbf{2.27\ V}$$

The emitter voltage is one diode drop less than the base voltage.

$$V_E = V_B - V_{BE} = 2.27\ \text{V} - 0.7\ \text{V} = \mathbf{1.57\ V}$$

Next, find the emitter current from Ohm's law.

$$I_E = \frac{V_E}{R_E} = \frac{1.57\ \text{V}}{470\ \Omega} = \mathbf{3.34\ mA}$$

Using the approximation $I_C \cong I_E$,

$$I_C = \mathbf{3.34\ mA}$$

Now find the collector voltage.

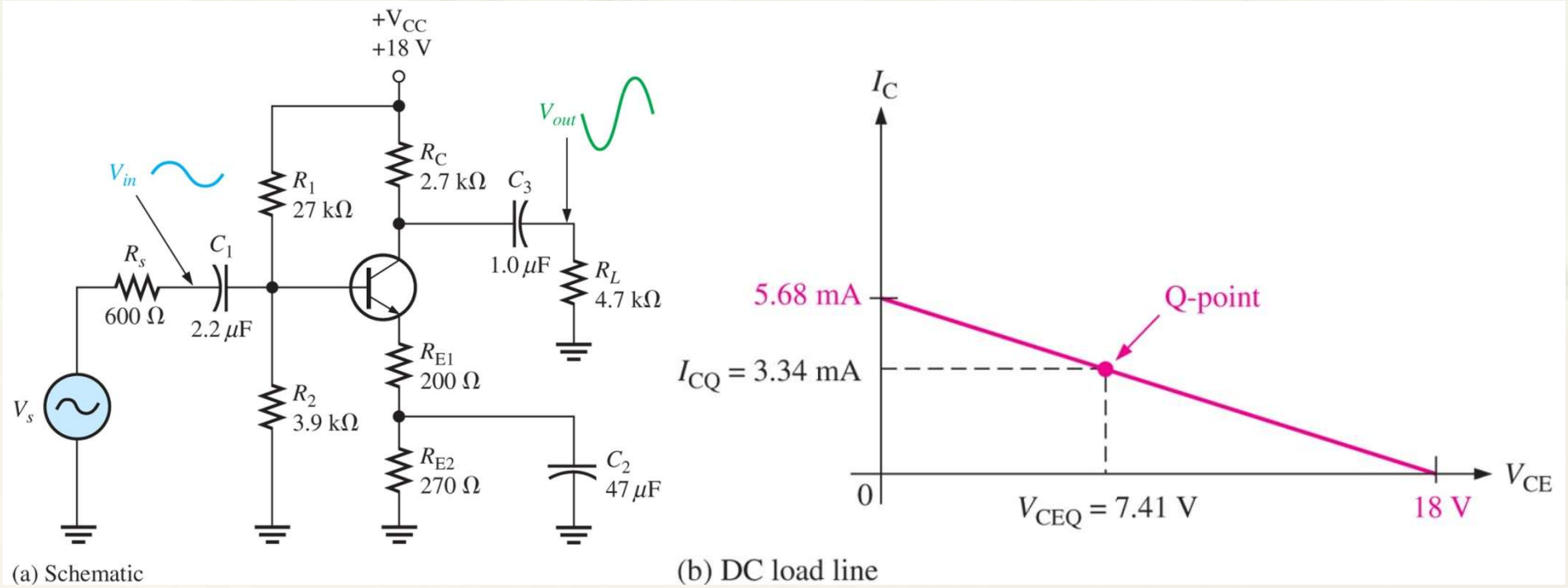
$$V_C = V_{CC} - I_C R_C = 18\ \text{V} - (3.34\ \text{mA})(2.7\ \text{k}\Omega) = 8.98\ \text{V}$$

The collector-emitter voltage is

$$V_{CE} = V_C - V_E = 8.98\ \text{V} - 1.57\ \text{V} = \mathbf{7.41\ V}$$

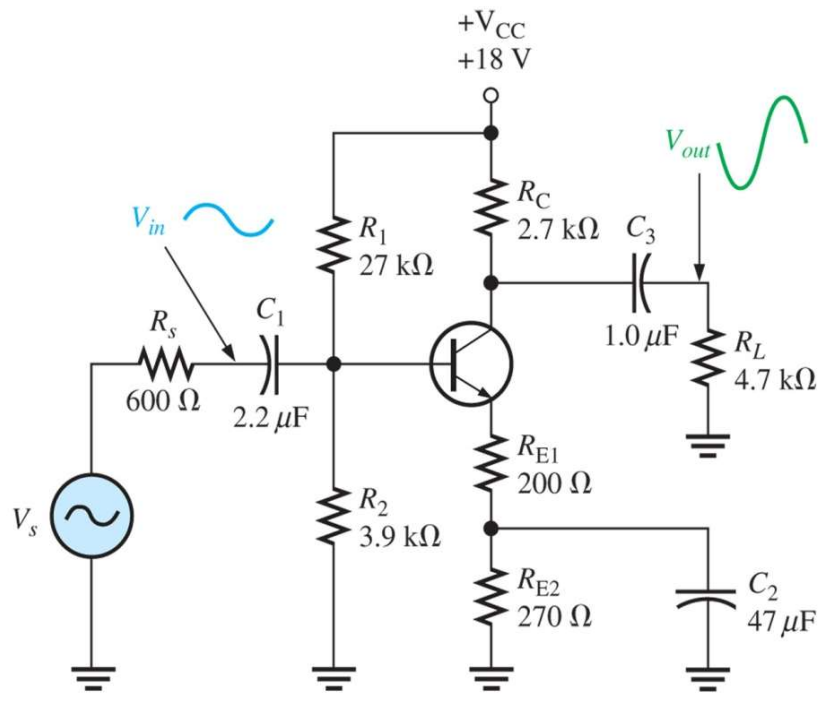
The Common-Emitter Amplifier

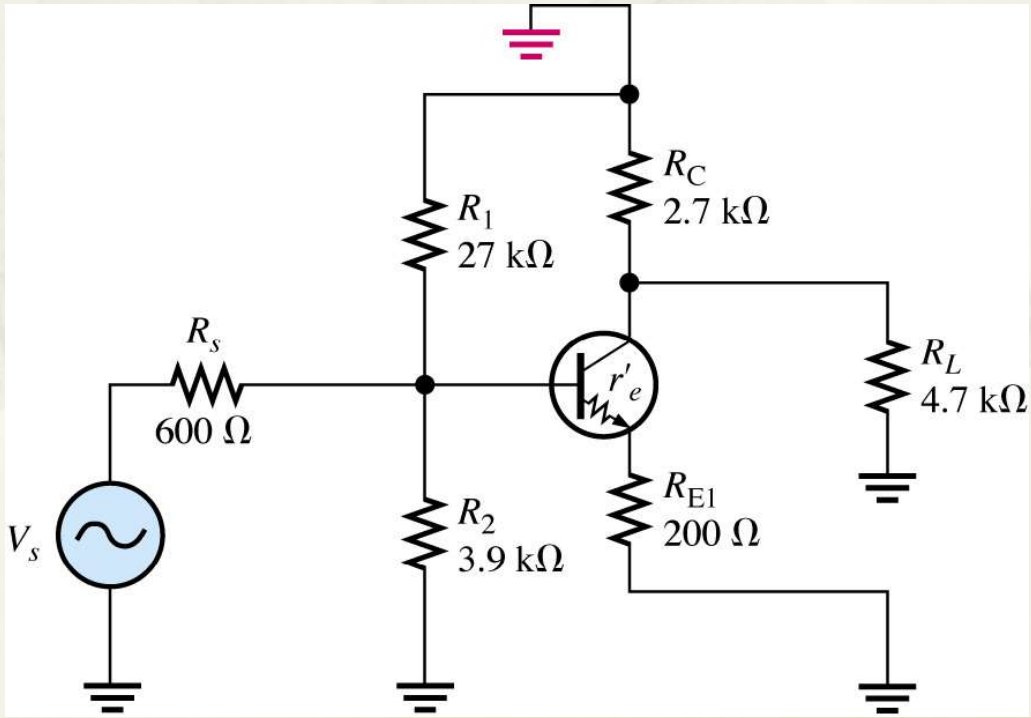
- **Determine the dc load line**
  - By finding the saturation current and cutoff voltage for the circuit
  - $V_{CE(cutoff)} = 18V, I_{C(sat)} = \frac{V_{CC}}{R_C+R_{E1}+R_{E2}} = \frac{18V}{2.7k\Omega+200\Omega+270\Omega} = 5.68mA$



The Common-Emitter Amplifier

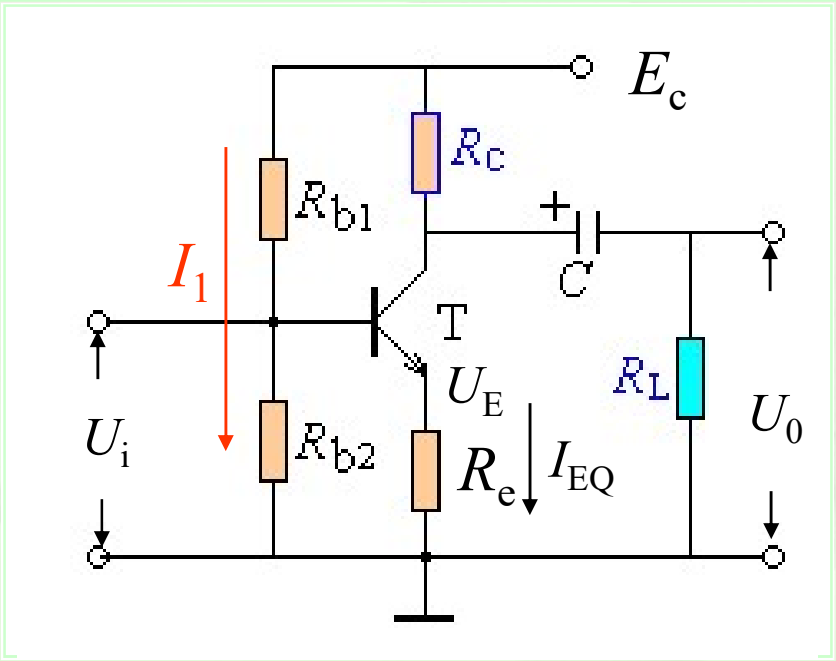
- **The ac equivalent circuit**
- **Dynamic emitter resistance (r_e')** $r_e' = \dfrac{25\ \text{mV}}{I_\text{E}}$

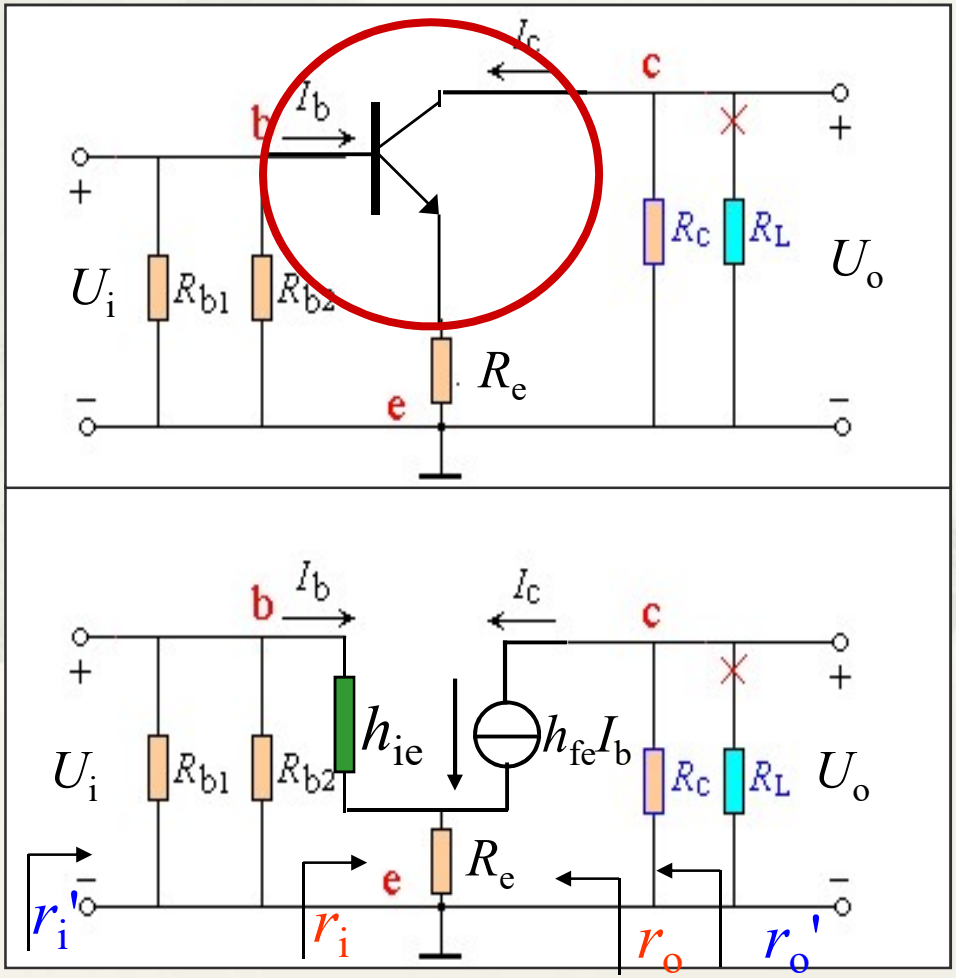




The Common-Emitter Amplifier

- **The ac equivalent circuit** – more intuitive





Example 3-8

Problem

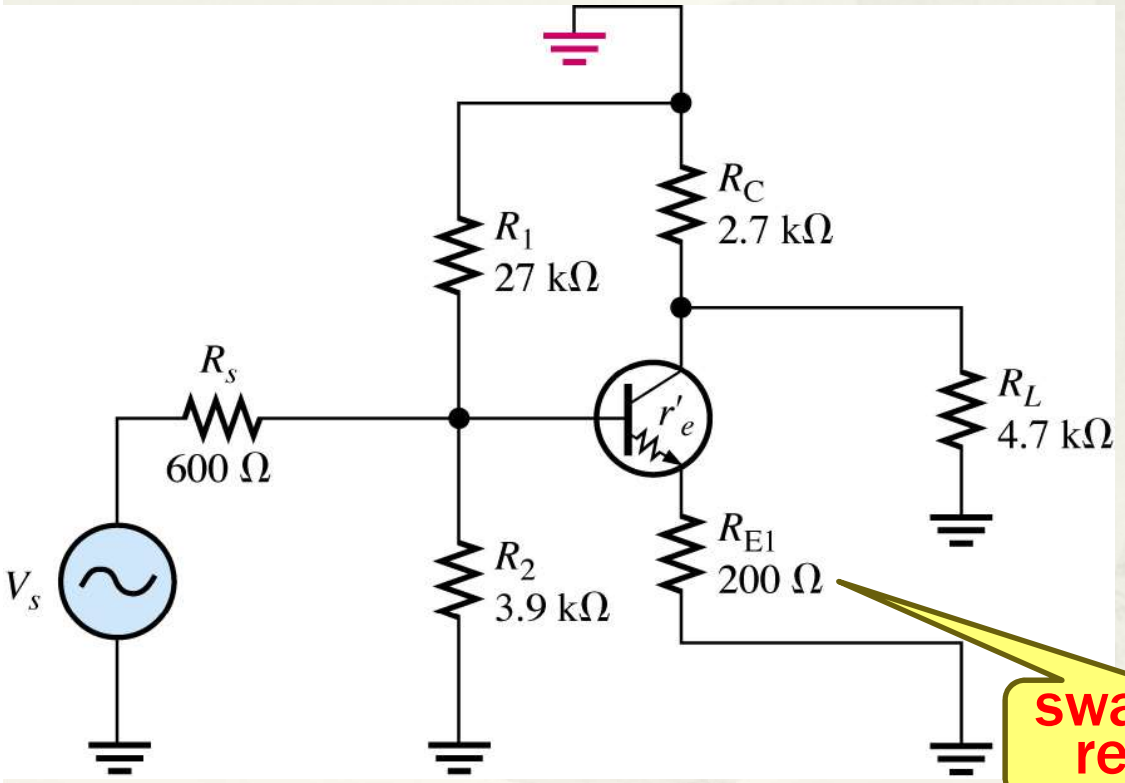
Find the dynamic emitter resistance, r'_e, for the circuit in Figure 3–28(a).

Solution

The emitter current was found to be 3.34 mA (see Example 3–4). Substituting into Equation 3–10,

$$r'_e = \frac{25\text{ mV}}{I_E} = \frac{25\text{ mV}}{3.34\text{ mA}} = \mathbf{7.5\ \Omega}$$

The Common-Emitter Amplifier



- **Voltage gain**
  - $A_v = \frac{v_{out}}{v_{in}} = \frac{v_c}{v_b}$

$$\cong \frac{v_c}{v_e} \qquad (v_b = v_e)$$

$$= -\frac{i_c R_c}{i_e R_e} \qquad (R_c = R_C \,//\, R_L)$$

$$\cong -\frac{R_c}{R_e} \qquad (i_c \cong i_e)$$

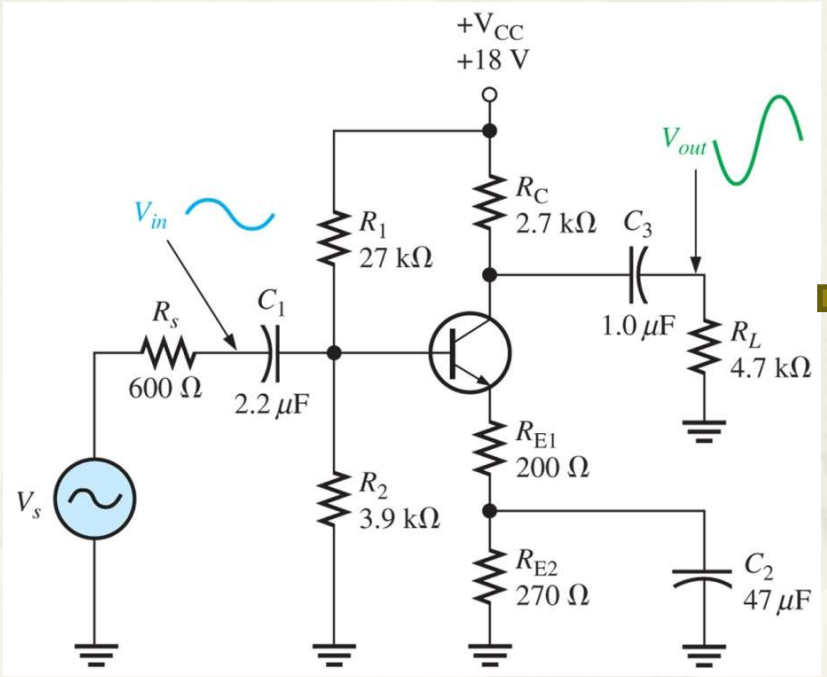
$$A_v = -\frac{R_c}{R_e}$$

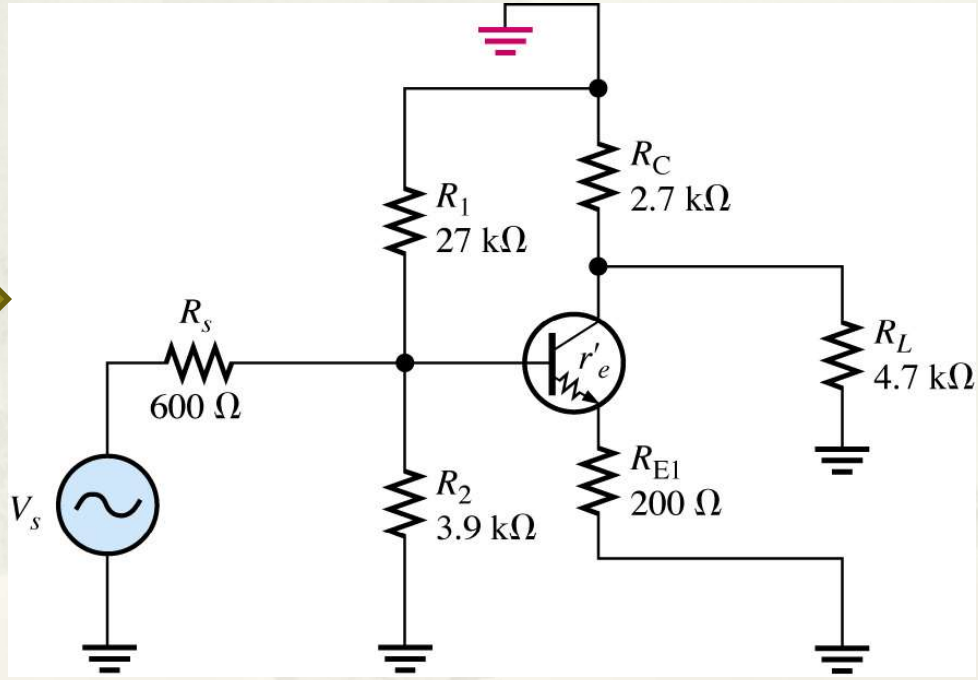
The collector and emitter resistances used in calculating gain are the total ac resistances in their respective circuits.

Example 3-9

Problem

Find A_v for the circuit in Figure 3–28(a).





Example 3-9

Solution

The ac resistance in the emitter circuit, R_e, is composed of r'_e in series with the unbypassed R_{E1}. From Example 3–8, $r'_e = 7.5\ \Omega$. Therefore,

$$R_e = r'_e + R_{E1} = 7.5\ \Omega + 200\ \Omega = 207.5\ \Omega$$

Next, find the ac resistance as viewed from the transistor's collector.

$$R_c = R_C \parallel R_L = 2.7\ \text{k}\Omega \parallel 4.7\ \text{k}\Omega = 1.71\ \text{k}\Omega$$

total ac resistance

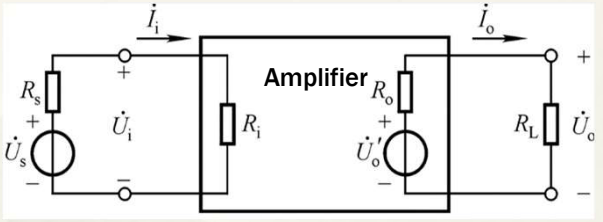
Substituting into Equation 3–11,

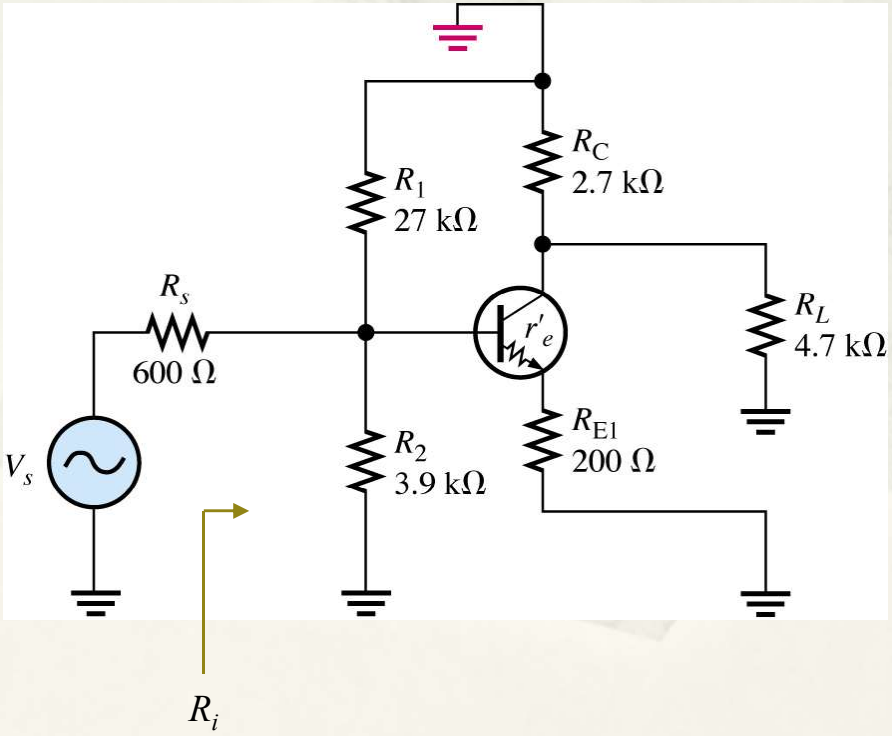
$$A_v = -\frac{R_c}{R_e} = -\frac{1.71\ \text{k}\Omega}{207.5\ \Omega} = \mathbf{-8.3}$$

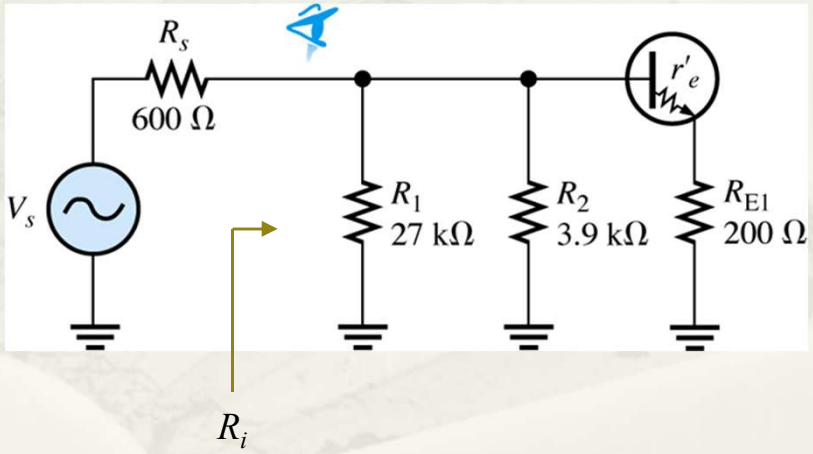
Again, the negative sign is used to show that the amplifier inverts the signal.

The Common-Emitter Amplifier

- **Input resistance ($R_{in(tot)}$)**
  - An ac parameter that acts like a load in series with the source resistance







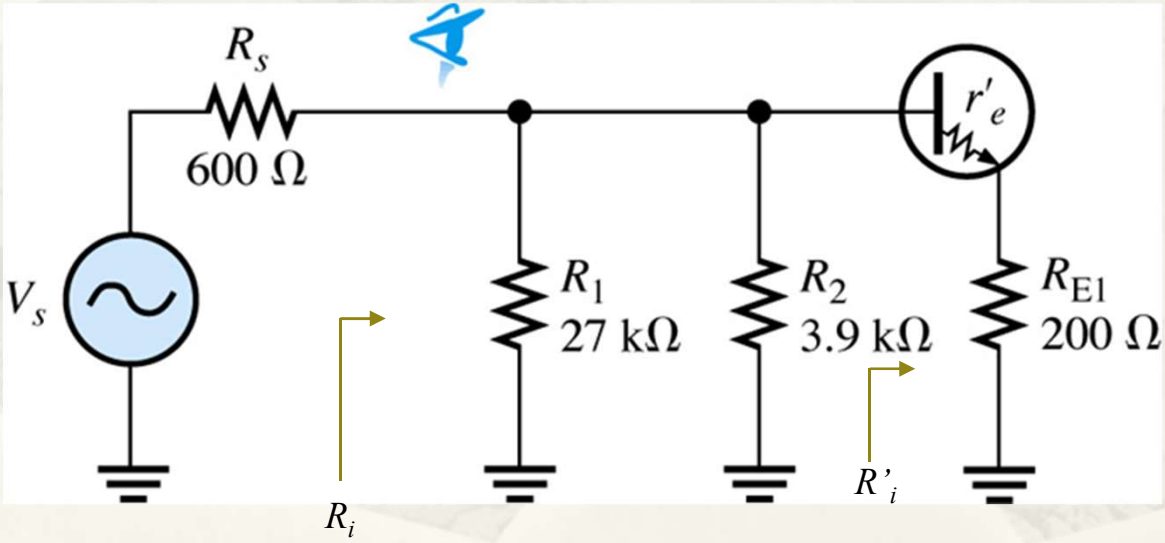
The Common-Emitter Amplifier

- **Input resistance ($R_{in(tot)}$)**
  - An ac parameter that acts like a load in series with the source resistance
  - $R_{in(tot)} = R_1 || R_2 || [\beta_{ac}(r'_e + R_{E1})]$

Amplifier

- Input resistance ($R_{in(tot)}$)
  - An ac parameter that acts like a load in series with the source resistance
  - $R_{in(tot)} = R_1 || R_2 || [\beta_{ac}(r'_e + R_{E1})]$

Dependent on β_{ac}



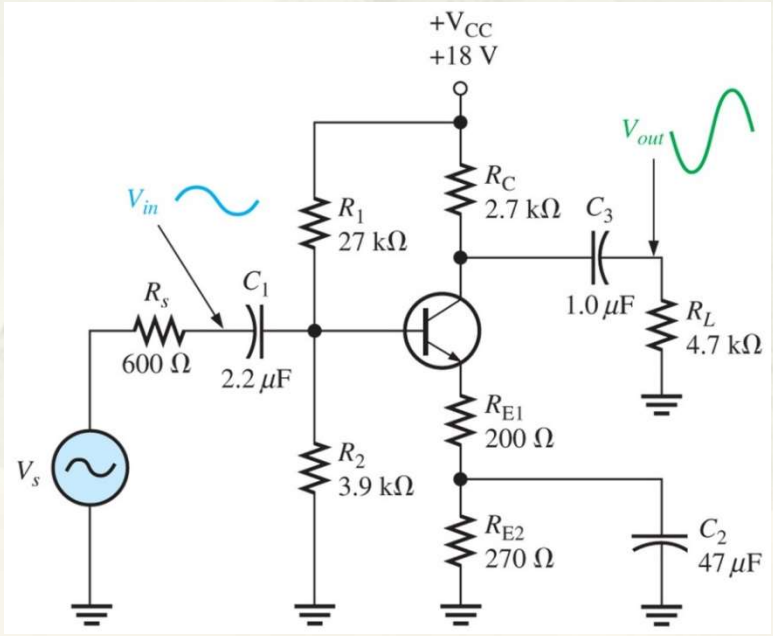
$$R'_i = \frac{v_b}{i_b} = \frac{(r'_e + R_{E1})i_b}{i_b} = r'_e + R_{E1}$$

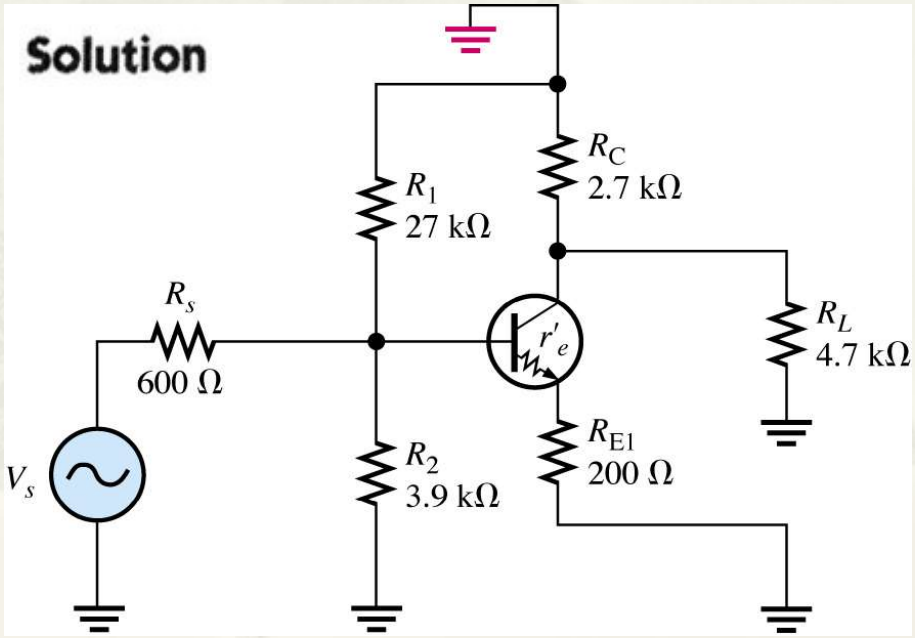
- Minimize the effect of β_{ac} and increase the total input resistance by adding a swamping resistorto the emitter circuit.

Example 3-10

Problem

Find $R_{in(tot)}$ for the circuit in Figure 3–28(a). Assume the β_{ac} is 120.



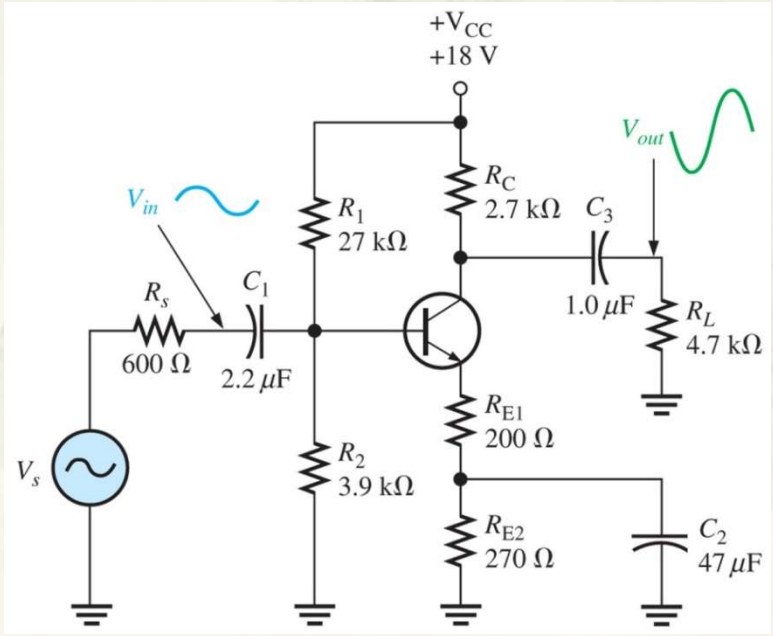


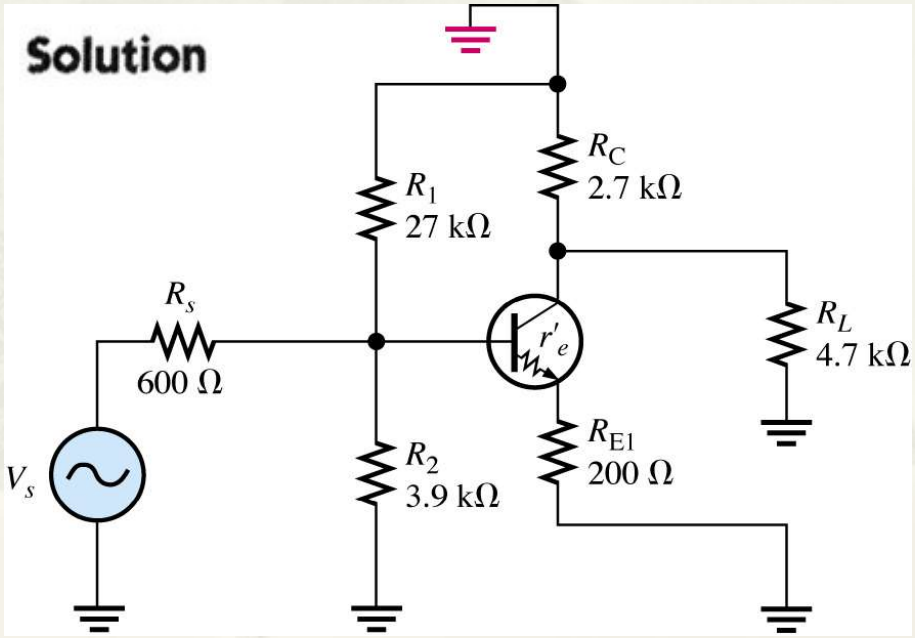
AC Equivalent Circuit

Example 3-10

Problem

Find $R_{in(tot)}$ for the circuit in Figure 3–28(a). Assume the β_{ac} is 120.





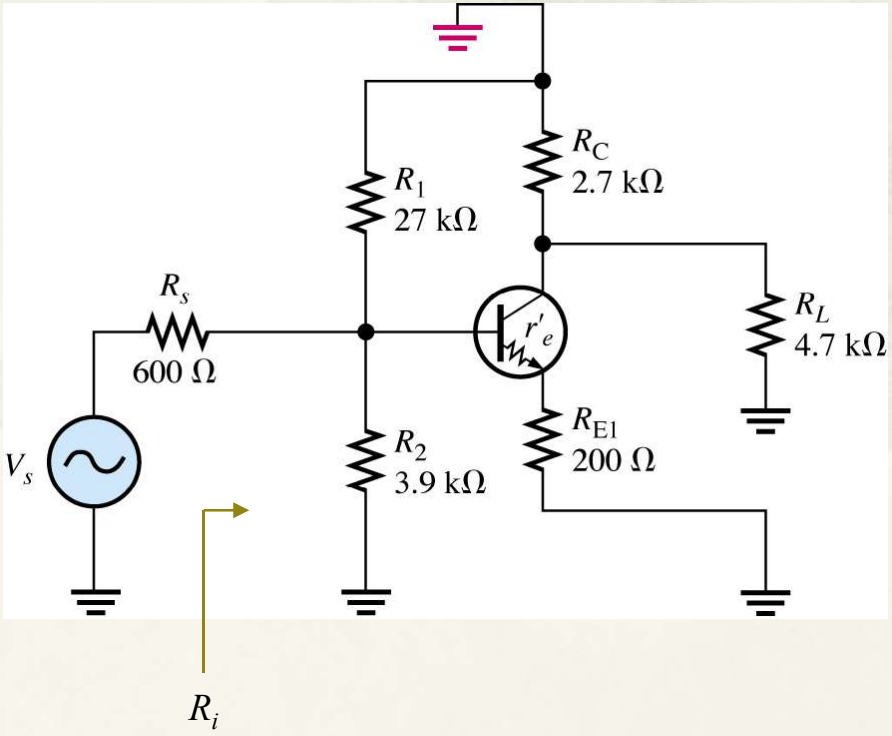
AC Equivalent Circuit

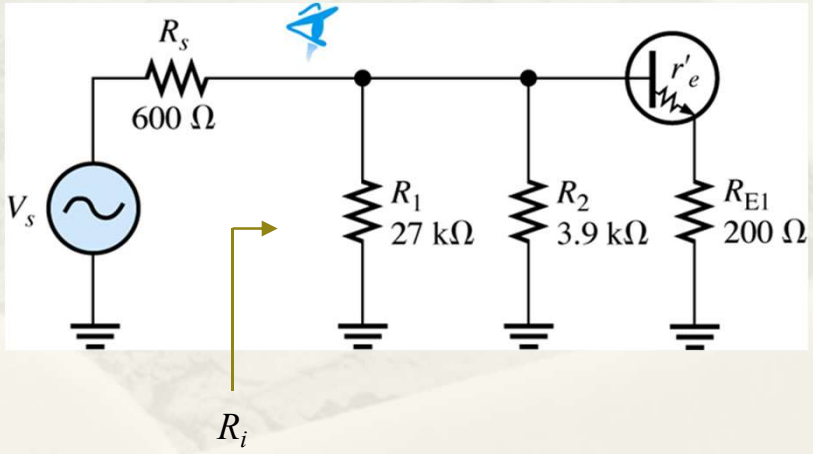
Example 3-10

Problem

Find $R_{in(tot)}$ for the circuit in Figure 3–28(a). Assume the β_{ac} is 120.

Solution





Example 3-10

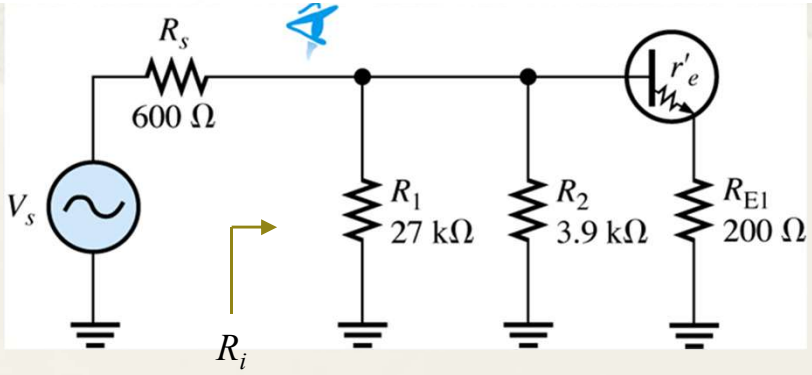
Problem

Find $R_{in(tot)}$ for the circuit in Figure 3–28(a). Assume the β_{ac} is 120.

Solution

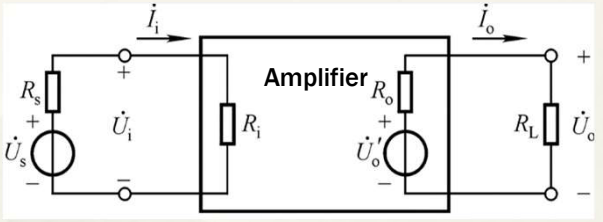
The internal ac emitter resistance, r'_e, was found to be 7.5 Ω in Example 3–8. Substituting into Equation 3–12,

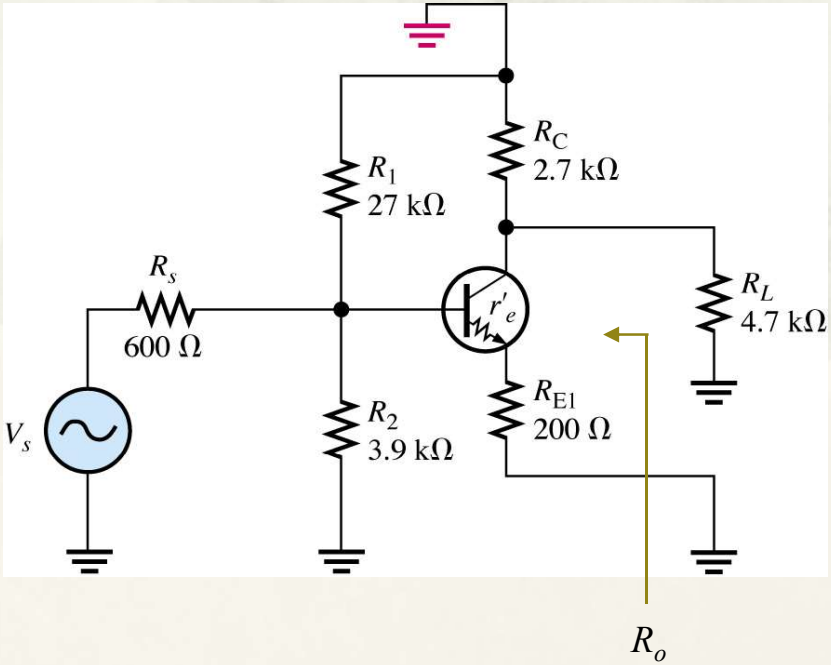
$$R_{in(tot)} = R_1 \parallel R_2 \parallel [\beta_{ac}(r'_e + R_{E1})]$$
$$= 27\ \text{k}\Omega \parallel 3.9\ \text{k}\Omega \parallel [120(7.5\ \Omega + 200\ \Omega)] = \mathbf{3.0\ k\Omega}$$

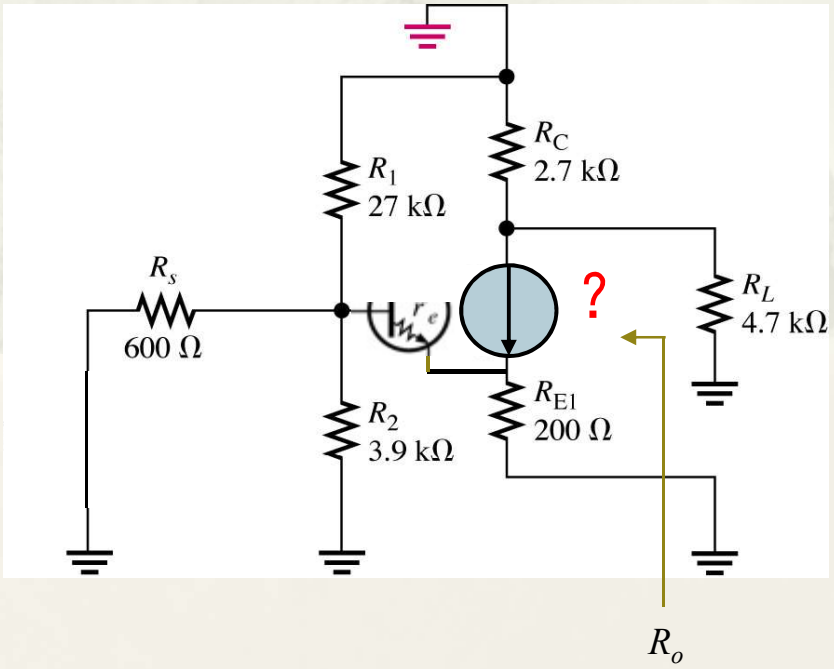


The Common-Emitter Amplifier

- **Output resistance (R_{out})**
  - An ac parameter that acts like a source resistor in series with the load resistance

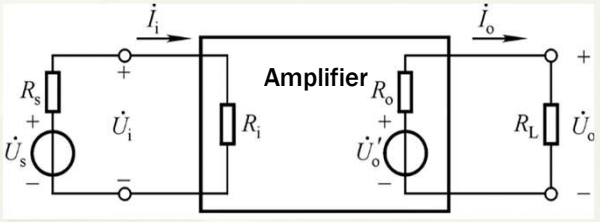


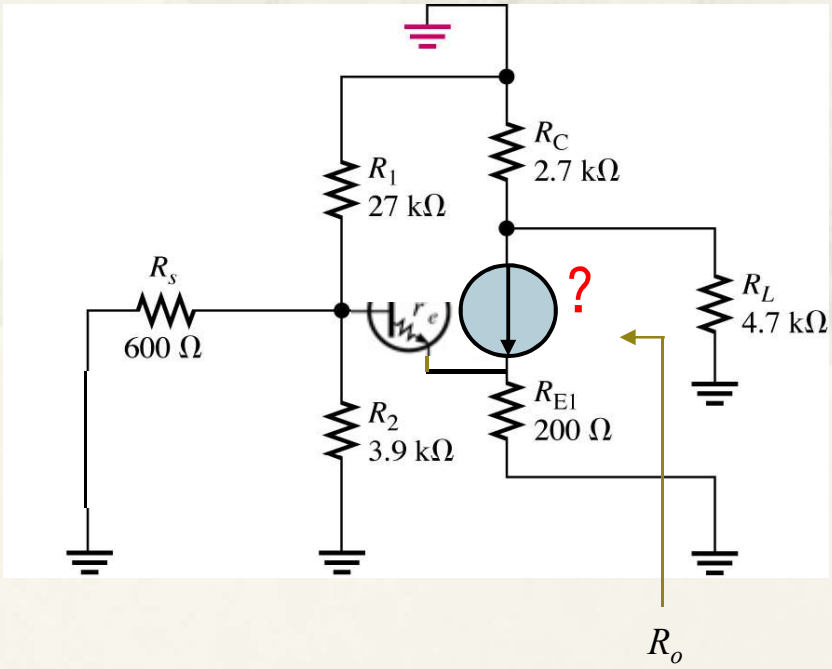


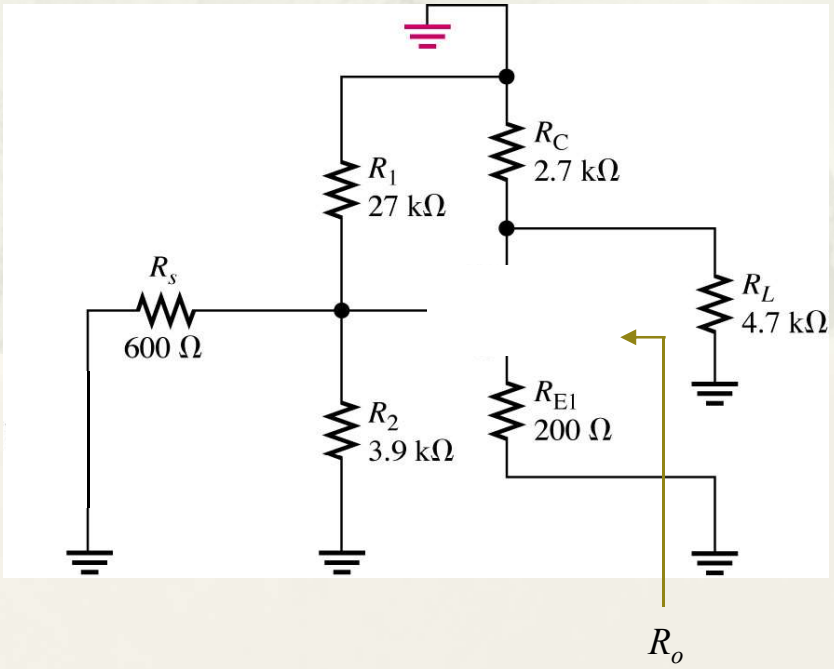


The Common-Emitter Amplifier

- **Output resistance (R_{out})**
  - An ac parameter that acts like a source resistor in series with the load resistance

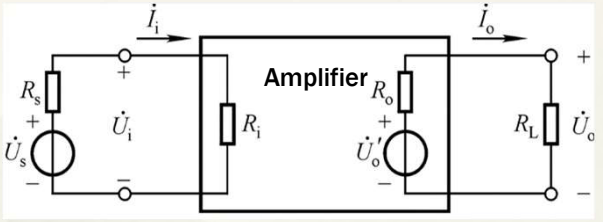


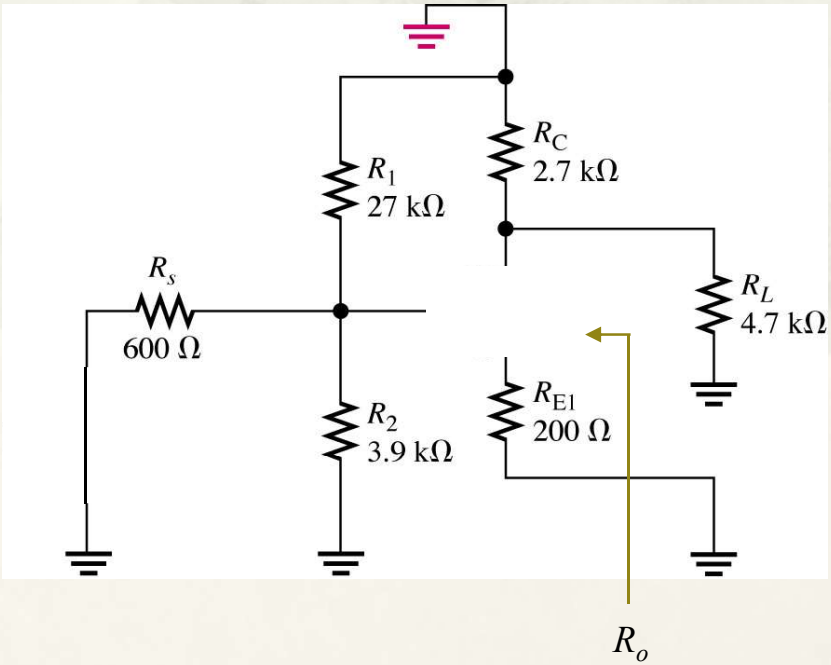


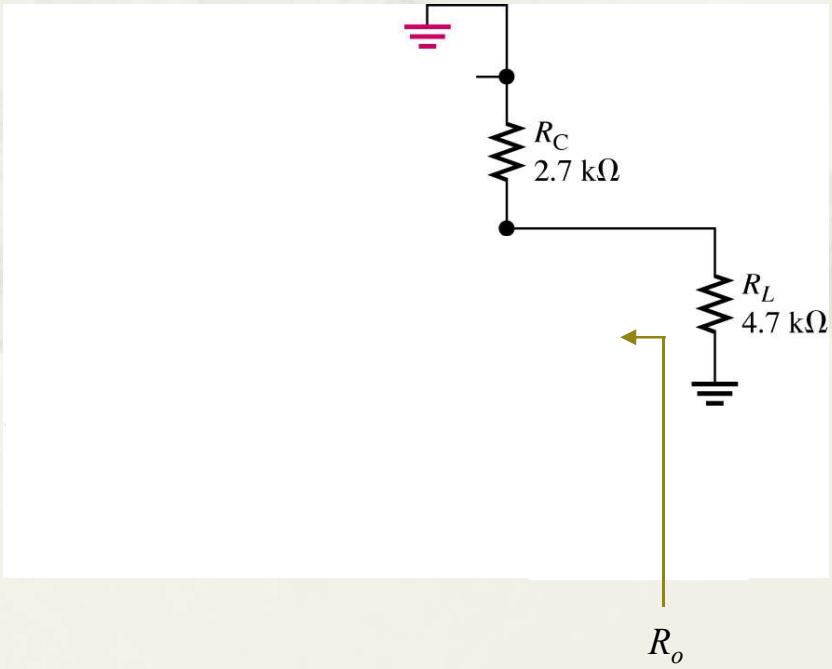


The Common-Emitter Amplifier

- **Output resistance (R_{out})**
  - An ac parameter that acts like a source resistor in series with the load resistance

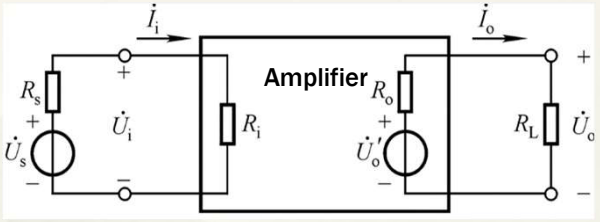


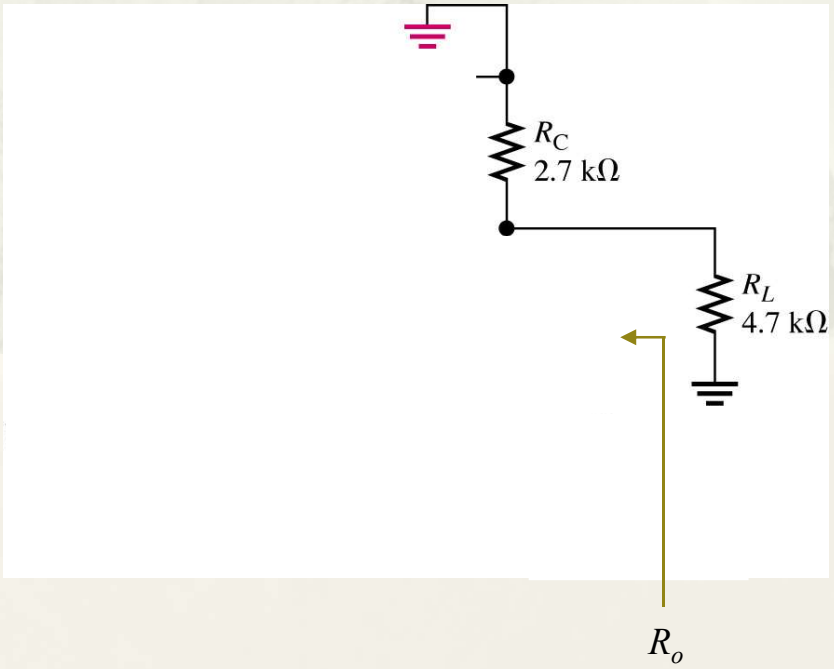




The Common-Emitter Amplifier

- **Output resistance (R_{out})**
  - An ac parameter that acts like a source resistor in series with the load resistance
  - $R_{out} = R_C$

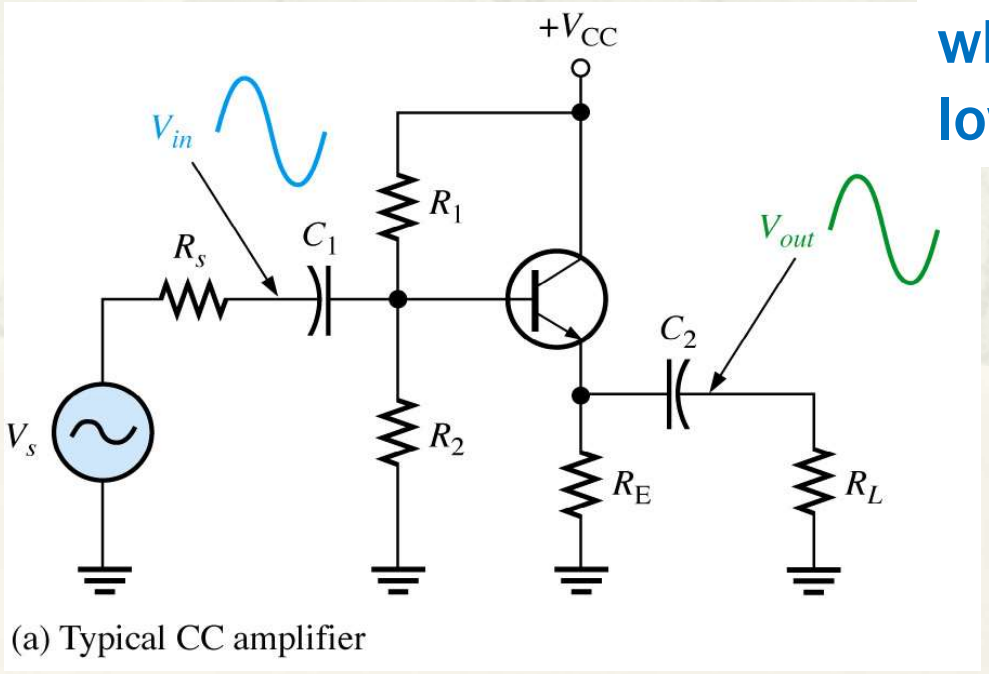




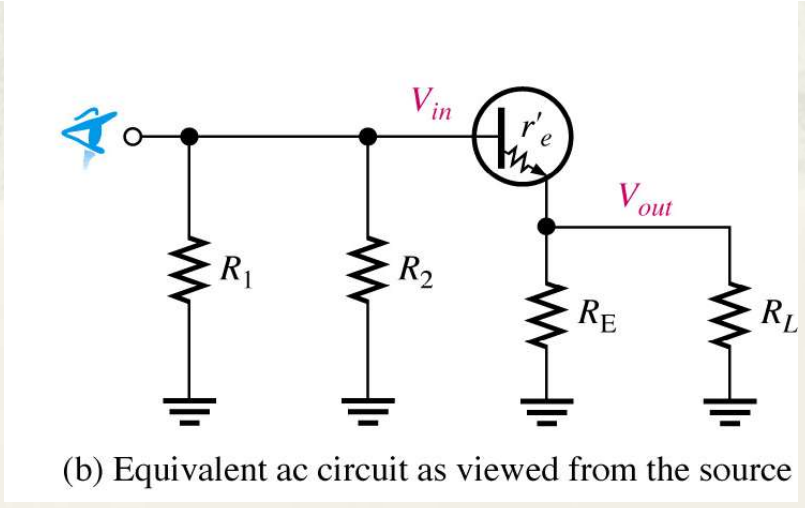
Common-Collector Amplifier

- **Common-collector (CC) amplifier (emitter-follower)**
  - Voltage gain :$A_v \cong 1$
  - Input resistor :$R_{in(tot)} = R_1 || R_2 || [\beta_{ac}(r'_e + R_{E1} || R_L)]$
  - Output resistor : $R_{out} \cong r'_e$

Useful in small power amplifiers when it is necessary to drive a low-impedance load.



(a) Typical CC amplifier

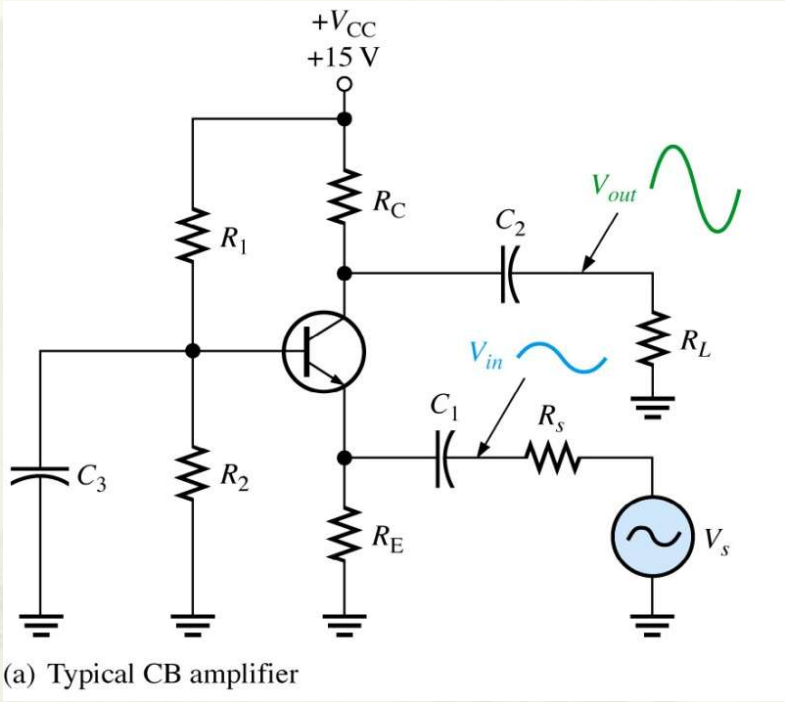


(b) Equivalent ac circuit as viewed from the source

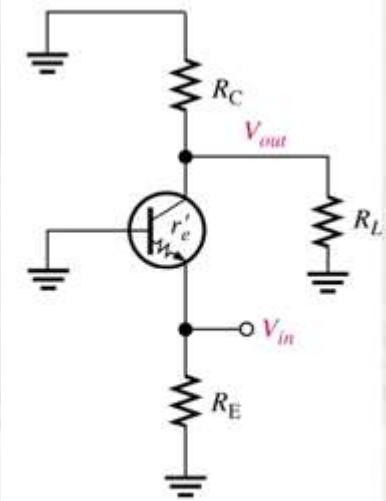
Common-Base Amplifier

- **Common-base (CB) amplifier**
  - Voltage gain :$A_v \cong \frac{R_c}{R_e}$
  - Input resistor :$R_{in(tot)} = r'_e || R_{\mathrm{E}}$
  - Output resistor : $R_{out} \cong R_c$

High frequency response than CE amplifiers



(a) Typical CB amplifier



(b)Equivalent ac circuit

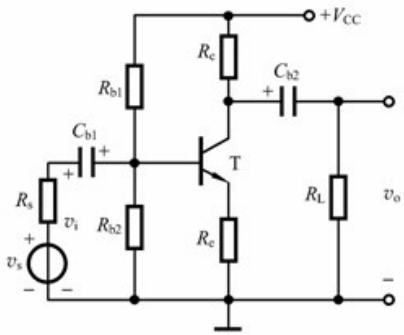
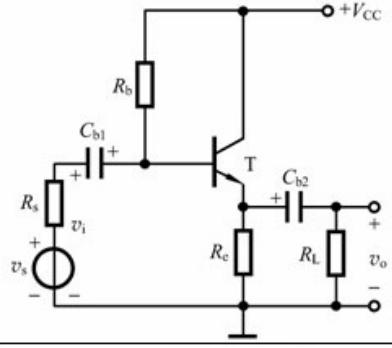
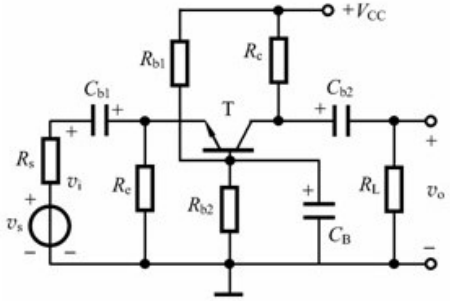
Comparison of Amplifier AC Parameters

TABLE 3–2

Comparison of amplifier ac parameters. Voltage-divider bias is assumed for all amplifiers with an unbypassed emitter resistor in the CE and CB configurations.

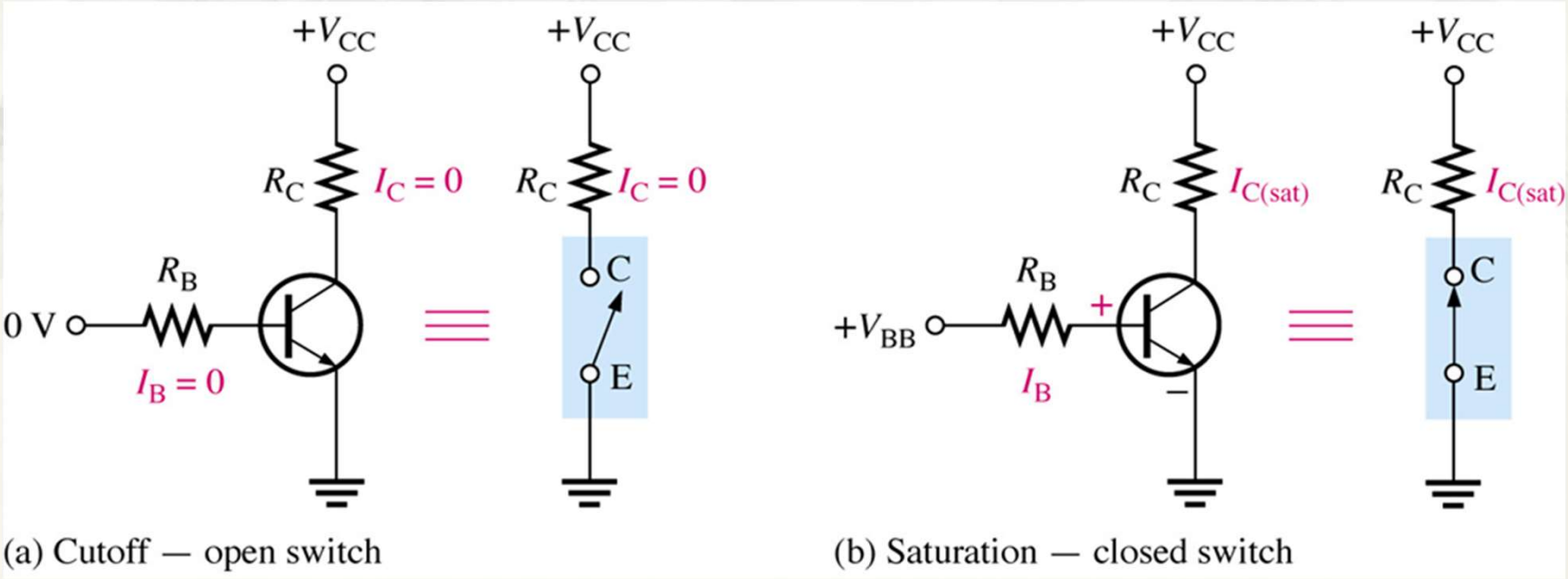
	CE	CC	CB
Voltage gain	$A_v = -\frac{R_c}{R_e}$ High	$A_v \cong 1$ Low	$A_v = \frac{R_c}{R_e}$ High
Input resistance	$R_{in(tot)} = R_1 \parallel R_2 \parallel [\beta_{ac}(r'_e + R_{E1})]$ Medium	$R_{in(tot)} = R_1 \parallel R_2 [\beta_{ac}(r'_e + R_E \parallel R_L)]$ High	$R_{in(tot)} = r'_e + R_{E1}$ Very low
Output resistance	R_C High	$\cong r'_e$ Low	R_C High
Phase relationship between V_{in} and V_{out}	180° inverted	0° not inverted	0° not inverted

Comparison of Amplifier AC Parameters

	共射极电路	共集电极电路	共基极电路
电路图			
电压增益 A_v	$A_v = -\dfrac{\beta R'_{\mathrm{L}}}{r_{\mathrm{be}} + (1+\beta)R_{\mathrm{e}}}$ $(R'_{\mathrm{L}} = R_{\mathrm{c}} \parallel R_{\mathrm{L}})$	$A_v = \dfrac{(1+\beta)R'_{\mathrm{L}}}{r_{\mathrm{be}} + (1+\beta)R'_{\mathrm{L}}}$ $(R'_{\mathrm{L}} = R_{\mathrm{e}} \parallel R_{\mathrm{L}})$	$A_v = \dfrac{\beta R'_{L}}{r_{\mathrm{be}}}$ $(R'_{L} = R_{\mathrm{e}} \parallel R_{L})$
v_o 与 v_i 的相位关系	反相	同相	同相
最大电流增益 A_i	$A_i \approx \beta$	$A_i \approx 1+\beta$	$A_i \approx \alpha$
输入电阻	$R_i = R_{b1} \parallel R_{b2} \parallel [r_{be} + (1+\beta)R_e]$	$R_i = R_b \parallel [r_{be} + (1+\beta)R'_L]$	$R_i = R_e \parallel \dfrac{r_{be}}{1+\beta}$
输出电阻	$R_o \approx R_c$	$R_o = \dfrac{r_{be} + R'_s}{1+\beta} \parallel R_e$ $(R'_s = R_s \parallel R_b)$	$R_o \approx R_c$
用　途	多级放大电路的中间级	输入级、中间级、输出级	高频或宽频带电路

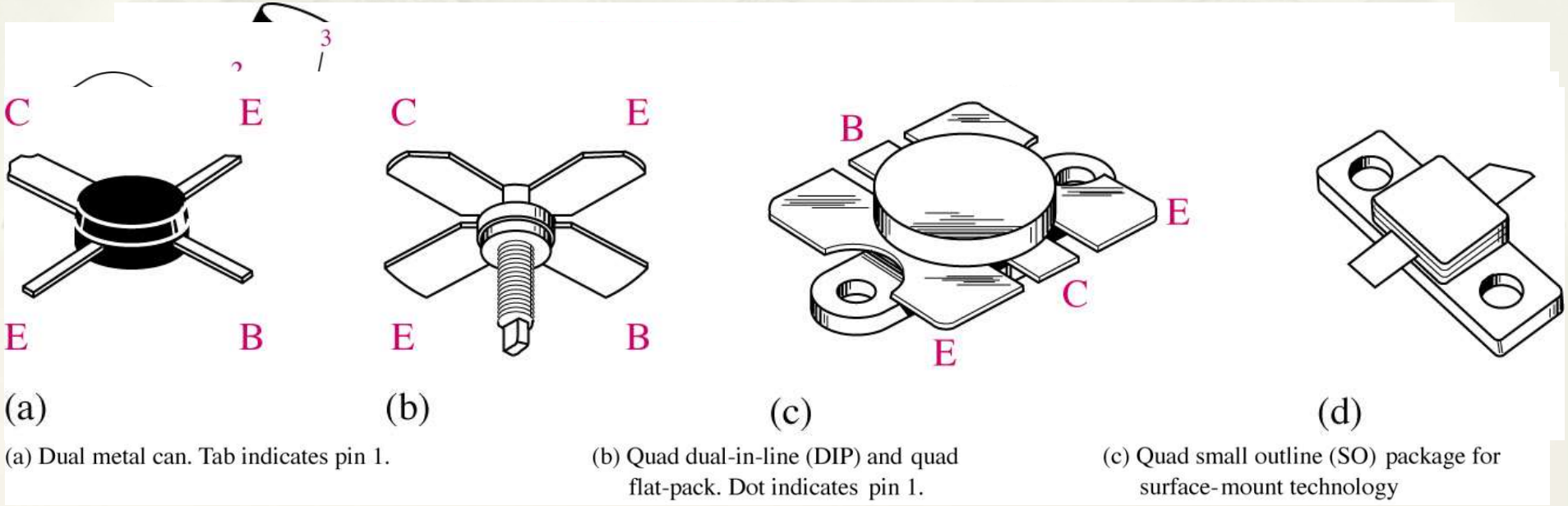
3-5 THE BIPOLAR TRANSISTOR AS A SWITCH

- **Digital circuit & Transistor switching circuit**
- **Ideal switching action of a transistor**
  - Cutoff, $I_C = 0, V_{CE} = V_{CC}, I_B = 0$: Open
  - Saturation, $I_C = \frac{V_{CC}}{R_C}, V_{CE} \cong 0, I_B \gg \frac{I_{C(sat)}}{\beta_{DC}}$: Closed



3-6 Transistor Packages and Terminal Identification

- **Transistor Categories**
  - General-purpose/Small-signal Transistors
  - Power Transistors
  - RF Transistors



(a) Dual metal can. Tab indicates pin 1.

(b) Quad dual-in-line (DIP) and quad flat-pack. Dot indicates pin 1.

(c) Quad small outline (SO) package for surface-mount technology

KEY OBJECTIVES

1. Describe the structure and operation of bipolar junction transistor (BJTs)
2. Explain how to bias a BJT with any of three types of biasing
3. Define basic amplifier terms and describe ac considerations for amplifiers
4. Analyze the operation of a small-signal amplifier
5. Describe how BJTs are used in switching circuits
6. Identify various packages used for transistors

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 3, 6, 8, 9, 11, 13, 19, 20, 21
- **Basic-plus Problems**
  - 27